

Omega Signature v0.1: Viable Propagation, Quotients, and the First Toy Witness

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Abstract

Omega is proposed as a formal program for studying viable propagation: the capacity of future-bearing structure to continue through time under constraints, perturbation, abstraction, and interaction. The intended long-term object is a universal, or at least quotient-stable, signature of such propagation:

$$\Omega_T(S) \approx \mathcal{P}_T^{\kappa^*}(S), \quad \kappa^* = \text{minimal admissible predictive quotient.}$$

This draft does not claim that the universal object has been obtained. Instead, it defines the current work-in-progress witness:

$$\Omega_{\text{witness}} = \mathcal{P}_T^{\kappa_{\text{COM}}}(S_{\text{toy}}),$$

certified multi-step viable propagation through center-of-mass macro-fibers in a two-field toy substrate.

The paper introduces the motivation for Omega, explains why survival, reachability, and entropy are insufficient as primary definitions, and develops the multifield fiber language needed to track viable transport without erasing component structure. It then formalizes the current COM witness, summarizes the empirical probe sequence that isolated it, and documents the controls and failure modes that constrain its interpretation. The central result is bounded: in the F, T attractive toy substrate, the COM quotient exposes a threshold-stable, null-separated, component-preserving propagation signature. The main unresolved issue is the quotient problem: whether this witness can be replaced by, generalized to, or embedded within a minimal admissible predictive quotient or quotient family.

This artifact is therefore both an entry point into the Omega theory and a status report on its first concrete toy signature. The broader aim remains a general theory of viable propagation; the current evidence supports a quotient-relative witness of the kind that theory is meant to explain.

Claim Ladder

- C1.** The Omega program aims at a universal or quotient-stable formal signature of viable propagation. *Status: theoretical objective.*
- C2.** In the toy multifield substrate, the center-of-mass quotient exposes positive certified multi-step viable propagation relative to product and shuffled null baselines. *Status: supported in Probes 09–12.*
- C3.** The COM witness is stable under threshold variation and substantial perturbation coverage. *Status: supported in Probe 10 and Probe 12A.*

- C4. Generic learned predictive quotients recover the COM witness. *Status: not supported; Probes 11–12 show partial signal but insufficient recovery, with fragmentation and fiber splitting/merging.*
- C5. The current COM witness generalizes to a quotient-independent or quotient-stable Omega object. *Status: open.*
- C6. Omega has direct implications for alignment, ecology, sociology, or value theory. *Status: speculative; outside the evidential scope of this artifact.*

1 Introduction

1.1 Motivation

The Omega program begins from a simple intuition: some systems do more than survive. They preserve and propagate the conditions under which viable futures remain reachable. A system may persist in a narrow sense while losing recoverability, collapsing into an irreversible trap, erasing lower-rank structure, or producing many apparent futures that do not actually propagate. Omega is intended to formalize the difference.

The central target is therefore not reward, utility, survival probability, or raw entropy. The target is viable propagation: the continued transport of future-bearing structure through time under constraints. In a mature theory, Omega would identify when a system keeps viable future structure reachable through perturbation, abstraction, and interaction.

This artifact reports the current state of that program. The strongest current result is not a universal theorem, but a bounded toy-substrate witness. In a two-field stochastic substrate, the F, T attractive corridor exhibits certified multi-step viable propagation through center-of-mass macro-fibers. This witness is separated from product and shuffled null baselines and from several tempting failure modes, including entropy-only structure and fragmented learned quotients.

1.2 From Entropy to Propagation

Early versions of the project used entropy over distinguishable viable futures as a tractable proxy. That proxy was useful, but the probe sequence exposed its limits. Entropy can be inflated by bad quotients, overfine partitions, or macro-labels that multiply apparent distinctions without preserving viable transport. A system can have many distinguishable futures and still fail to carry viable structure forward.

The current formulation therefore demotes entropy. Entropy remains a breadth diagnostic, but only after viability and propagation have been established. The core question is not how many futures are distinguishable, but whether viable structure propagates through certified channels over time.

1.3 Why a Toy Witness Matters

The current COM witness matters because it makes the abstract target concrete. It defines an explicit quotient, explicit fibers, explicit transport edges, explicit certification rules, explicit null baselines, and explicit failure modes. This turns Omega from a broad philosophical proposal into a testable mathematical object, albeit in a deliberately limited toy setting.

The toy witness should be interpreted as a proof of concept, not as a final theory. It shows that the kind of structure Omega seeks can be isolated, measured, and distinguished from weaker

alternatives in at least one controlled substrate. The next question is whether the witness can be formalized, reproduced, generalized, or replaced by a learned admissible quotient.

1.4 Contributions

This artifact makes six contributions:

1. It states the long-term aim of Omega as a universal or quotient-stable signature of viable propagation.
2. It distinguishes that aim from the current evidential status: a quotient-relative COM fiber-transport witness in a toy multifield substrate.
3. It summarizes the single-field lessons that led to a propagation-first formulation and the demotion of entropy.
4. It defines the multifield COM fiber-transport construction in terms of fibers, certified nodes, certified edges, certified paths, component preservation, and null-baseline separation.
5. It reports the current empirical witness from Probes 09–12 and summarizes why competing or learned quotients have not yet replaced COM.
6. It identifies the quotient problem, generalization problem, and formalization problem as the main open directions.

2 Ambition and Evidential Status

The long-term aim of the Omega program is a universal formal object: a description-independent, or at least quotient-stable, signature of viable propagation. In its strongest intended form, Omega would characterize when future-bearing structure can continue to propagate through constraints, perturbations, abstraction, and interaction.

The present artifact does not claim that this universal object has been obtained. It reports a bounded work-in-progress result: in a toy multifield substrate, a center-of-mass quotient exposes certified multi-step viable propagation through macro-fibers, separated from product and shuffled null baselines and from several failure modes.

This distinction is central. The theoretical foreground remains ambitious: Omega is intended to become a general theory of viable propagation. The current evidential status is narrower: we have a quotient-relative toy witness. The gap between these two levels is not a defect to be hidden, but the research program itself.

The quotient problem is therefore load-bearing. Current evidence supports

$$\Omega_{\text{witness}} = \mathcal{P}_T^{\kappa_{\text{COM}}}, \quad (1)$$

not yet

$$\Omega_T = \mathcal{P}_T^{\kappa^*}. \quad (2)$$

Here κ_{COM} is a hand-defined center-of-mass quotient, while κ^* denotes the intended minimal admissible predictive quotient. The former is the current witness; the latter remains an open target.

If a quotient-stable κ^* exists, Omega may become close to a system property. If no such quotient exists, the appropriate object may instead be a family of description-relative signatures:

$$\{\mathcal{P}_T^\kappa(S) : \kappa \in \mathcal{K}_{\text{adm}}\}. \quad (3)$$

That possibility is not treated as a failure of the present result. It is one of the main questions the present result makes precise. The COM construction shows that at least one nontrivial quotient exposes robust viable propagation in the toy substrate. Future work must determine whether such quotients can be characterized, learned, or shown to form a stable equivalence class.

Thus the claim of this artifact is deliberately bounded: the COM construction is a concrete witness of the kind of viable propagation structure the universal theory is meant to capture. It is not the final theory.

3 The Intended Final Object

3.1 Omega as Viable Propagation

The intended object of the Omega program is not a scalar reward, a utility function, a survival probability, or an entropy maximum. It is a formal signature of viable propagation: the capacity of future-bearing structure to continue through time under constraints.

In the strongest version of the theory, Omega would identify whether a system preserves the conditions under which viable futures remain reachable, recoverable, and transportable. This motivates a propagation-first ordering:

1. viability;
2. certified transport;
3. multi-step propagation depth;
4. recoverability;
5. component preservation;
6. nonfragmentation;
7. breadth or entropy of what propagates.

The ordering matters. Breadth is meaningful only after viable propagation has been established. A large set of distinguishable futures that does not propagate is not Omega-positive.

3.2 Quotient-Indexed Propagation

Let S be a system or state space, let T be a horizon, and let κ be a macro-map or quotient. For a given quotient, the operational object is:

$$\mathcal{P}_T^\kappa(S), \tag{4}$$

where $\mathcal{P}_T^\kappa$ denotes certified viable propagation through the macro-structure induced by κ .

This quotient-indexed form is the currently safest formal level. It makes explicit that propagation is measured relative to a descriptive map. The toy results support this kind of object:

$$\Omega_{\text{witness}} = \mathcal{P}_T^{\kappa_{\text{COM}}}(S_{\text{toy}}). \tag{5}$$

Here S_{toy} is the toy multifield substrate and κ_{COM} is the center-of-mass quotient.

3.3 The Universal Target

The larger theoretical ambition is stronger. Ideally, there exists a minimal admissible predictive quotient κ^* such that Omega can be written as:

$$\Omega_T(S) = \mathcal{P}_T^{\kappa^*}(S). \quad (6)$$

The quotient κ^* would not be arbitrary. It would be the simplest macro-description that preserves predictive sufficiency for future viable propagation. Informally, two microstates or trajectory segments should be identified by κ^* when they have the same relevant future propagation profile.

One candidate characterization is:

$$\kappa^* = \arg \min_{\kappa \in \mathcal{K}} [\mathcal{L}_{\text{prop}}(\kappa) + \lambda \mathcal{L}_{\text{quotient}}(\kappa)], \quad (7)$$

where $\mathcal{L}_{\text{prop}}$ measures loss of predictive sufficiency for future viable propagation, and $\mathcal{L}_{\text{quotient}}$ penalizes unnecessary quotient complexity, fragmentation, and degenerate label structure.

This object remains aspirational. Probes 11–12 do not recover such a κ^* . They instead show that simple predictive quotients can see part of the signal but fail to reproduce the current COM witness.

3.4 Admissible Quotient Families

If no unique κ^* exists, the appropriate object may be a constrained family of admissible descriptions rather than a single quotient:

$$\Omega_T(S) \rightsquigarrow \{\mathcal{P}_T^\kappa(S) : \kappa \in \mathcal{K}_{\text{adm}}\}. \quad (8)$$

Here $\mathcal{K}_{\text{adm}}$ is a family of quotients satisfying admissibility constraints such as predictive sufficiency, nonfragmentation, component preservation, and robustness under perturbation or abstraction.

This fallback is important. It allows Omega to remain meaningful even if description-independent uniqueness fails. In that case, Omega would not be a single observer-free scalar attached to the microdynamics alone. It would be a structured family of viable-propagation signatures under admissible descriptions.

The current COM result should be understood in this light. It is evidence that the toy substrate admits at least one admissible-looking quotient under which nontrivial viable propagation is certified. It is not evidence that the admissible quotient family has been fully characterized.

3.5 Current Witness Versus Final Object

The distinction between the current witness and the intended final object is:

$$\text{current supported object:} \quad \Omega_{\text{witness}} = \mathcal{P}_T^{\kappa_{\text{COM}}}(S_{\text{toy}}), \quad (9)$$

$$\text{intended universal object:} \quad \Omega_T(S) = \mathcal{P}_T^{\kappa^*}(S), \quad (10)$$

$$\text{description-family fallback:} \quad \Omega_T(S) = \{\mathcal{P}_T^\kappa(S) : \kappa \in \mathcal{K}_{\text{adm}}\}. \quad (11)$$

Only the first is presently supported by the experiments. The second is the long-term theoretical target. The third is the conservative fallback if the quotient problem turns out to be inherently description-relative.

This paper is therefore not a claim that universal Omega has been found. It is a claim that one quotient-relative witness of the intended type has been isolated, audited, and separated from several failure modes in a controlled toy substrate.

4 Formal Primitives

This section introduces the minimal formal vocabulary used throughout the artifact. The definitions are intentionally generic. The COM witness introduced later is one specific instantiation of these primitives in the toy multifield substrate.

4.1 States, Dynamics, and Trajectories

Let S denote a state space. In the multifield setting, the joint state space is written:

$$S^{AB} = S^A \times S^B, \quad (12)$$

where S^A and S^B are the component state spaces.

A joint state at time t is:

$$s_t^{AB} = (s_t^A, s_t^B). \quad (13)$$

A trajectory over horizon T is:

$$\tau_{0:T}^{AB} = (s_0^{AB}, s_1^{AB}, \dots, s_T^{AB}). \quad (14)$$

The transition law is denoted abstractly by P . In the toy substrate, P contains the stochastic single-field dynamics and the interaction term between fields A and B . The details of P are implementation-specific; the formal construction below requires only sampled or computed trajectories and a viability predicate.

4.2 Viability

Let V be a viability predicate on trajectories or trajectory prefixes. In the strict horizon- T case:

$$V(\tau_{0:T}) \in \{0, 1\}. \quad (15)$$

A trajectory is viable over horizon T if:

$$V(\tau_{0:T}) = 1. \quad (16)$$

The viable trajectory set from initial state s_0 is:

$$\mathcal{F}_T(s_0) = \{\tau_{0:T} : \tau_0 = s_0, V(\tau_{0:T}) = 1\}. \quad (17)$$

For the multifield case:

$$\mathcal{F}_T^{AB}(s_0^{AB}) = \{\tau_{0:T}^{AB} : \tau_0^{AB} = s_0^{AB}, V(\tau_{0:T}^{AB}) = 1\}. \quad (18)$$

Viability is a gate. Nonviable trajectories do not contribute to Omega propagation except as failure modes, collapse modes, or null comparisons.

4.3 Quotients

A quotient or macro-map is a function:

$$\kappa : S \rightarrow Z, \quad (19)$$

where Z is a set of macro-labels.

In a trajectory-segment implementation, κ may instead act on a local trajectory window:

$$\kappa : \tau_{t:t+\Delta} \rightarrow Z. \quad (20)$$

The macro-label at time or segment boundary t_i is:

$$z_i = \kappa(s_{t_i}). \quad (21)$$

A quotient is not assumed to be unique or canonical. The quotient problem is one of the main open issues of the Omega program: which macro-maps are admissible, and whether there exists a minimal predictive quotient κ^* .

4.4 Fibers

Given a quotient κ , a macro-label z , and a time or segment boundary t_i , the viable fiber over z is:

$$\mathcal{R}_{t_i}^\kappa(z) = \{\tau_{0:T} \in \mathcal{F}_T : \kappa(s_{t_i}) = z\}. \quad (22)$$

In the multifield setting:

$$\mathcal{R}_{t_i}^{\kappa, AB}(z) = \{\tau_{0:T}^{AB} \in \mathcal{F}_T^{AB} : \kappa(s_{t_i}^{AB}) = z\}. \quad (23)$$

A fiber represents the set of viable micro-realizations supporting a given macro-label. The fiber language is central because Omega propagation should not be credited to unsupported or one-off macro-labels.

4.5 Fiber Mass and Nonfragmentation

Let μ denote the empirical or theoretical measure over viable trajectories. The viable fiber mass of z at t_i is:

$$M_{t_i}^\kappa(z) = \mu(\mathcal{R}_{t_i}^\kappa(z)). \quad (24)$$

A quotient can fail by overfragmenting the trajectory set. The simplest diagnostic is singleton fraction:

$$\text{singleton_fraction} = \frac{\#\{z : |\kappa^{-1}(z)| = 1\}}{\#\{z\}}. \quad (25)$$

A related diagnostic is small-fiber fraction:

$$\text{small_fiber_fraction} = \frac{\#\{z : |\kappa^{-1}(z)| < m\}}{\#\{z\}}, \quad (26)$$

where m is a fixed small-fiber threshold.

These diagnostics are not cosmetic. They test whether the macro-map is really abstracting a supported structure or merely re-labeling microscopic detail.

4.6 Transport Edges

For adjacent segment boundaries t_i and t_{i+1} , a transport edge is an observed macro-transition:

$$z_i \rightarrow z_{i+1}. \quad (27)$$

The transported viable mass of this edge is:

$$M_i^\kappa(z_i, z_{i+1}) = \mu \left(\left\{ \tau_{0:T} \in \mathcal{F}_T : \kappa(s_{t_i}) = z_i, \kappa(s_{t_{i+1}}) = z_{i+1} \right\} \right). \quad (28)$$

Transport edges are the local carriers of viable propagation. A high count of macro-labels is not sufficient; viable mass must be transported between supported fibers over time.

4.7 Certified Nodes and Edges

A macro-node is certified when its fiber has sufficient viable support and passes nonfragmentation and component-preservation checks. Abstractly:

$$\text{CertNode}(z, t_i; \kappa) = 1 \quad (29)$$

if the node satisfies the certification criteria.

A macro-edge is certified when it has sufficient transported viable mass and connects certified nodes:

$$\text{CertEdge}(z_i, z_{i+1}, t_i; \kappa) = 1. \quad (30)$$

The specific thresholds used in the current toy witness are given in Section 8 and the operational appendix.

4.8 Certified Paths

A certified macro-path is a sequence:

$$\gamma = (z_0, z_1, \dots, z_K) \quad (31)$$

such that each node and each adjacent edge is certified:

$$\text{CertNode}(z_i, t_i; \kappa) = 1 \quad \text{for all } i, \quad (32)$$

$$\text{CertEdge}(z_i, z_{i+1}, t_i; \kappa) = 1 \quad \text{for all } i < K. \quad (33)$$

Let Γ_T^κ denote the set of certified macro-paths over horizon T .

The mass of a certified path is:

$$M^\kappa(\gamma) = \mu \left(\left\{ \tau_{0:T} \in \mathcal{F}_T : \kappa(s_{t_i}) = z_i \text{ for each } i \right\} \right). \quad (34)$$

4.9 Viable Propagation

The path survival mass is:

$$S_{\text{path}}^\kappa = \sum_{\gamma \in \Gamma_T^\kappa} M^\kappa(\gamma). \quad (35)$$

The multi-step transport depth, denoted $D_{\text{transport}}^\kappa$, is a summary of how far certified viable transport persists across segment boundaries. Its implementation-specific form is reported in the operational appendix; conceptually, it measures whether propagation survives across many steps rather than appearing only as a local transition.

The quotient-indexed viable propagation object is then written:

$$\mathcal{P}_T^\kappa = \text{Propagate}(\mathcal{F}_T, \kappa, \Gamma_T^\kappa), \quad (36)$$

where the right-hand side denotes the structured tuple of viable mass, certified path survival, multi-step depth, component preservation, nonfragmentation, and null-baseline separation.

4.10 Null Baselines

The current probes use two primary null baselines.

The product baseline estimates what propagation would look like if the two fields composed independently. The shuffled baseline preserves marginal trajectory information while destroying coupling-specific joint structure.

For any propagation statistic $\mathcal{P}_T^\kappa$, define:

$$\Delta_{\text{prod}}^\kappa = \mathcal{P}_{T,\text{coupled}}^\kappa - \mathcal{P}_{T,\text{product}}^\kappa, \quad (37)$$

$$\Delta_{\text{shuf}}^\kappa = \mathcal{P}_{T,\text{coupled}}^\kappa - \mathcal{P}_{T,\text{shuffled}}^\kappa. \quad (38)$$

Positive null separation is not sufficient by itself, but it is required for a coupled multifield witness.

4.11 Component Projections

In the multifield setting, a joint fiber should not count as Omega-like if it preserves only an aggregate while erasing one component. Let:

$$\pi_A : S^{AB} \rightarrow S^A, \quad \pi_B : S^{AB} \rightarrow S^B \quad (39)$$

be the component projections.

For a joint fiber $\mathcal{R}_{t_i}^{\kappa, AB}(z)$, the projected component fibers are:

$$\pi_A \left(\mathcal{R}_{t_i}^{\kappa, AB}(z) \right), \quad (40)$$

$$\pi_B \left(\mathcal{R}_{t_i}^{\kappa, AB}(z) \right). \quad (41)$$

Component preservation scores C_A and C_B measure whether component structure remains supported along certified nodes and paths. Lower-rank erasure tracks collapse of component support inside a joint propagation channel.

The exact implementation-level scores are operational diagnostics. Their conceptual role is fixed: a multifield witness must propagate joint structure without erasing the lower-rank structures that realize it.

5 Single-Field Omega: Status and Lessons

The multifield COM witness did not appear in isolation. It follows from a sequence of single-field tests that clarified what Omega could and could not be. The most important lesson from the single-field program is that Omega cannot be identified with survival, raw reachability, or entropy alone. The object must track viable propagation: whether future-bearing structure continues through time in a way that is recoverable, nondegenerate, and robust under admissible abstraction.

5.1 The Original Single-Field Proxy

The earliest tractable single-field observable was based on the entropy of distinguishable viable futures. For an initial state s and horizon T , define the viable future set:

$$\mathcal{F}_T(s) = \{\tau_{0:T} : \tau_0 = s, V(\tau_{0:T}) = 1\}. \quad (42)$$

Given a coarse-graining or quotient C , the quotient future set is:

$$\tilde{\mathcal{F}}_T^C(s) = \mathcal{F}_T(s)/C. \quad (43)$$

The initial proxy was:

$$\mathcal{I}_T^C(s) = H\left(\tilde{\mathcal{F}}_T^C(s)\right). \quad (44)$$

This proxy was useful because it separated survival from future richness. A state with a single surviving path should not be treated the same as a state with many distinguishable viable continuations. However, the proxy also exposed a problem: entropy is highly quotient-dependent. A bad quotient can collapse meaningful structure, while an overfine quotient can manufacture apparent richness.

5.2 Survival Is Not Enough

The first single-field lesson is that survival alone is too weak. A trajectory may avoid immediate failure while entering a rigid attractor, irreversible trap, or low-recoverability region. Such a trajectory persists in a narrow sense, but it does not preserve future-bearing structure.

In Omega terms, survival is a gate:

$$V(\tau_{0:T}) = 1, \quad (45)$$

but survival is not the object. A viable trajectory contributes to Omega only when it participates in a structure that can continue to propagate.

5.3 Raw Reachability Is Not Enough

Raw reachability also fails. A state may reach many future states, but those futures may be nonviable, transient, unrecoverable, or generated by noise rather than coherent structure. Counting reachable futures without viability and propagation certification admits false positives.

This motivates the distinction:

$$\text{Reach}_T(s) \neq \mathcal{F}_T(s) \neq \mathcal{P}_T^\kappa(s). \quad (46)$$

Reachability asks what can be accessed. Viability asks what remains within the constraint set. Propagation asks what viable structure continues through time under an admissible description.

5.4 Entropy Is a Breadth Diagnostic, Not Omega

Entropy was demoted because it measures breadth, not propagation. Breadth is important only after viable propagation has already been established.

A quotient may produce high entropy by fragmenting the state space:

$$H(\kappa(s_t)) \uparrow \quad (47)$$

while failing to transport viable structure:

$$\mathcal{P}_T^\kappa \downarrow. \quad (48)$$

Conversely, a system may reduce raw breadth while improving coherent viable transport. This later became central in the multifield probes: the interesting object was not maximal joint entropy, but certified propagation through viable macro-fibers.

The revised status of entropy is therefore:

$$H_{\text{breadth}} = H(\text{propagated viable macro-classes}), \quad (49)$$

where entropy is measured only over classes that have already passed viability and propagation tests.

5.5 Irreversibility and Traps

Single-field tests also highlighted the importance of irreversibility. A region that leads to an absorbing sink, rigid trap, or unrecoverable attractor should reduce Omega even if it temporarily preserves viability or produces apparent branching.

Let $R_T(s)$ denote a recoverability or return-to-viability diagnostic. Then a state with high short-term viability but low recoverability should not be treated as Omega-rich:

$$p_{\text{viable}}(s) \text{ high but } R_T(s) \text{ low} \Rightarrow \text{propagation risk.} \quad (50)$$

This lesson later reappears in the multifield setting as lower-rank erasure and component collapse. A system can preserve a macro-aggregate while destroying the conditions that allow its components to continue.

5.6 Recoverability and Flexibility

Recoverability is the ability of viable structure to survive perturbation, repair, or local deviation. In single-field terms, recoverability distinguishes a merely surviving path from a path embedded in a robust viable neighborhood.

A recoverable future has nearby alternatives that can restore viability:

$$s_t \rightarrow s_{t+1} \text{ with perturbation } \delta s_t \Rightarrow \exists \text{ viable continuation.} \quad (51)$$

Recoverability therefore points toward the propagation-first view. A future is not Omega-relevant merely because it exists. It is Omega-relevant when viable structure can be carried forward despite local disturbance.

5.7 Horizon Coherence

Omega should not be a short-horizon artifact. A structure that appears viable at small T but collapses at larger T is not robustly propagating. This motivates horizon coherence: the qualitative propagation profile should remain coherent as T increases, or else its collapse should be treated as a meaningful failure mode.

For a quotient κ , horizon coherence asks whether:

$$\mathcal{P}_{T_1}^\kappa \quad \text{and} \quad \mathcal{P}_{T_2}^\kappa \tag{52}$$

remain structurally compatible for $T_2 > T_1$. The requirement is not that the numerical value remain constant. Rather, the certified propagation structure should not disappear merely because the horizon becomes long enough to reveal traps, fragmentation, or irreversible collapse.

5.8 The Admissibility Problem

The single-field work made the quotient problem unavoidable. The same dynamics can look rich, collapsed, or noisy depending on the coarse-graining. Therefore Omega cannot be defined as entropy or propagation under an arbitrary quotient.

A quotient should be admissible only if it preserves relevant future viable structure while avoiding trivial extremes:

- an all-one quotient collapses distinctions;
- an identity-like quotient overfits microscopic detail;
- a random or hash-like quotient can launder entropy;
- a useful quotient preserves predictive structure without excessive fragmentation.

This is the precursor to the multifield quotient problem. The current COM witness is quotient-relative precisely because admissibility is not yet solved in general.

5.9 Single-Field Status

The single-field program should be understood as having established a profile, not a final scalar. The relevant profile includes:

$$(p_{\text{viable}}, H_{\text{breadth}}, R_T, \text{irreversibility}, \text{horizon coherence}, \text{quotient admissibility}). \tag{53}$$

The main status claims are:

1. Single-field probes support the demotion of survival and entropy as primary definitions.
2. Viability remains a necessary gate.
3. Recoverability and horizon coherence are essential to propagation.
4. Quotient admissibility is load-bearing.

5. The single-field result motivates, but does not by itself solve, the multifield fiber-transport construction.

Thus, the single-field work did not finish Omega. It changed the target. It moved the project from “entropy of viable futures” toward “certified viable propagation under admissible quotient structure.” The multifield COM witness is the first concrete toy-substrate expression of that revised target.

Next I’d draft **“Why Multifield Omega Is Needed”**, since it bridges single-field lessons to the COM fiber witness.

““latex

6 Why Multifield Omega Is Needed

The single-field program clarifies viability, recoverability, horizon coherence, and quotient admissibility. However, many systems of interest are not single fields. They are coupled systems whose viability depends on interaction, composition, and preservation of lower-rank structure. Multifield Omega asks whether viable propagation survives when fields interact.

6.1 From Single-Field Propagation to Coupled Propagation

In the single-field setting, the basic question is:

Given a state, what viable structure can propagate forward from it?

In the multifield setting, the question becomes:

Given interacting fields, does viable structure compose, collapse, erase, fragment, or propagate through the coupled system?

Let A and B be two fields with component state spaces S^A and S^B . The joint state space is:

$$S^{AB} = S^A \times S^B. \quad (54)$$

A coupled trajectory is:

$$\tau_{0:T}^{AB} = ((s_0^A, s_0^B), (s_1^A, s_1^B), \dots, (s_T^A, s_T^B)). \quad (55)$$

The multifield viable set is:

$$\mathcal{F}_T^{AB} = \{\tau_{0:T}^{AB} : V^{AB}(\tau_{0:T}^{AB}) = 1\}. \quad (56)$$

The central issue is that $\mathcal{F}_T^{AB}$ generally does not reduce to a simple product of the single-field viable sets.

6.2 Nonfactorization Under Interaction

For independent fields, one might expect:

$$\mathcal{F}_T^{AB} \approx \mathcal{F}_T^A \times \mathcal{F}_T^B. \quad (57)$$

Under interaction, this approximation can fail:

$$\mathcal{F}_T^{AB} \neq \mathcal{F}_T^A \times \mathcal{F}_T^B. \quad (58)$$

Coupling can change the viable future structure in several ways. It can:

- expand accessible futures;
- compress accessible futures;
- stabilize otherwise fragile trajectories;
- introduce new traps or irreversible collapse modes;
- preserve a joint macro-structure while erasing component structure;
- create apparent breadth without viable propagation;
- enable transport through macro-fibers that does not exist in the product baseline.

This is why multifield Omega cannot be reduced to comparing single-field survival or entropy scores. The object of interest is the structure of viable propagation under coupling.

6.3 Why Product and Shuffled Baselines Matter

The multifield probes use null baselines to distinguish genuine coupled propagation from marginal artifacts.

The product baseline asks what would be expected if the two fields composed independently. It tests whether the joint signal exceeds what is already present in the separate fields.

The shuffled baseline preserves marginal trajectory information while breaking coupling-specific joint correlations. It tests whether apparent joint transport depends on real coupling structure or merely on the marginal distribution of component trajectories.

For a quotient-indexed propagation statistic $\mathcal{P}_T^\kappa$, the relevant deltas are:

$$\Delta_{\text{prod}}^\kappa = \mathcal{P}_{T,\text{coupled}}^\kappa - \mathcal{P}_{T,\text{product}}^\kappa, \quad (59)$$

$$\Delta_{\text{shuf}}^\kappa = \mathcal{P}_{T,\text{coupled}}^\kappa - \mathcal{P}_{T,\text{shuffled}}^\kappa. \quad (60)$$

A coupled multifield witness should show positive or otherwise interpretable separation from these baselines. Without such separation, the result may be explained by independent component behavior or by marginal trajectory structure.

6.4 Fibers as the Natural Multifield Language

In a multifield system, a macro-state is not sufficient by itself. We also need to know what lower-level realizations support it. This motivates the fiber language.

Given a quotient κ and a macro-label z , the fiber:

$$\mathcal{R}_t^\kappa(z) = \{\tau_{0:T}^{AB} \in \mathcal{F}_T^{AB} : \kappa(s_t^{AB}) = z\} \quad (61)$$

contains the viable joint trajectories realizing that macro-label at time or segment boundary t . The fiber asks:

What viable micro-realizations support this macro-state?

Transport between fibers asks:

Does the macro-state carry viable support forward through time?

This is the natural language for multifield Omega because it keeps the macro description tied to the viable lower-level realizations that make it real.

6.5 Component Preservation and Lower-Rank Erasure

A multifield witness must preserve more than a joint aggregate. It must avoid erasing the lower-rank structures that realize the joint state.

Let:

$$\pi_A : S^{AB} \rightarrow S^A, \quad \pi_B : S^{AB} \rightarrow S^B \quad (62)$$

be the component projections. A joint fiber $\mathcal{R}_t^{\kappa, AB}(z)$ induces projected component structures:

$$\pi_A \left(\mathcal{R}_t^{\kappa, AB}(z) \right), \quad (63)$$

$$\pi_B \left(\mathcal{R}_t^{\kappa, AB}(z) \right). \quad (64)$$

If a macro-channel preserves z while one component projection collapses, the channel is not Omega-like. It is a form of lower-rank erasure. This motivates component preservation scores C_A and C_B , along with a lower-rank erasure diagnostic E_{lower} .

The requirement is not that every microscopic distinction be preserved. That would collapse back into identity-like overfragmentation. The requirement is that the joint propagation channel remain supported by nontrivial viable component structure.

6.6 Compression Can Be Omega-Relevant

The multifield probes also changed the interpretation of breadth. A naive expectation might be that coupling should increase the number of viable joint futures. But the current evidence suggests a subtler possibility: coupling can compress raw future breadth while improving the coherence of viable transport.

This motivates the phrase:

not necessarily wider futures, but more carryable futures.

In formal terms, a coupled system may have lower raw breadth,

$$H_{\text{breadth,coupled}} < H_{\text{breadth,product}}, \quad (65)$$

while exhibiting stronger certified propagation,

$$\mathcal{P}_{T,\text{coupled}}^\kappa > \mathcal{P}_{T,\text{product}}^\kappa. \quad (66)$$

Such a regime is Omega-relevant because the object is viable propagation, not entropy maximization.

6.7 Why Multifield Is the Right Next Test

Multifield systems introduce exactly the issues that the single-field program could not resolve:

1. whether viable structure composes across fields;
2. whether coupling enables propagation unavailable to independent fields;
3. whether macro-stability hides component erasure;
4. whether quotient choice exposes or destroys transport structure;
5. whether apparent richness is genuine propagation or pseudo-risk.

The COM witness is the current answer in one controlled toy substrate. It shows that under the center-of-mass quotient, the F, T attractive corridor supports certified multi-step viable propagation through macro-fibers. The remainder of the paper formalizes that witness and its limitations.

7 Experimental Substrate

The empirical work in this artifact uses a deliberately small toy multifield substrate. The purpose of the substrate is not physical realism. Its purpose is to provide a controlled setting in which viability, coupling, traps, null baselines, quotient choice, and fiber transport can be tested explicitly.

The substrate should therefore be read as a validation environment for candidate formal objects, not as a model of any particular real-world system.

7.1 Two Coupled One-Dimensional Fields

The system consists of two stochastic one-dimensional fields, denoted A and B . A joint state is:

$$s_t^{AB} = (s_t^A, s_t^B), \quad (67)$$

where each component state includes at least a position coordinate:

$$x_t^A, \quad x_t^B. \quad (68)$$

The joint dynamics are sampled over a finite horizon T . The transition law contains both single-field dynamics and an interaction term. Abstractly:

$$s_{t+1}^{AB} \sim P_\alpha(s_t^{AB}), \quad (69)$$

where α is the coupling strength. In the main multifield probes, the coupling is attractive.

The implementation includes stochasticity, so all propagation statistics are estimated empirically from sampled trajectories and bootstrapped over seeds.

7.2 Viability and the Sink

The substrate includes an irreversible sink or failure region. A trajectory is viable over horizon T if it avoids the sink and satisfies the experiment’s viability predicate over the full trajectory:

$$V(\tau_{0:T}^{AB}) = 1. \quad (70)$$

The sink is important because it prevents viability from collapsing into raw reachability. A trajectory that reaches many states but eventually enters an irreversible sink is not counted as a viable propagated future.

This mirrors the single-field lesson: traps and irreversible collapse should reduce Omega-relevant propagation even when short-horizon activity appears rich.

7.3 The F, T Attractive Corridor

The current witness is isolated in a specific initial-pair corridor, denoted F, T . This notation refers to the initial qualitative regions or field conditions used in the toy substrate. The important point is not the name of the regions, but that the pair defines a reproducible initial condition class for the two-field system.

The main coupled condition is:

$$(F, T) \quad \text{with attractive coupling.} \quad (71)$$

This corridor was retained because earlier probes showed that it supports a nontrivial viable propagation signal under the center-of-mass quotient. Other initial pairs and coupling families are not treated as established in this artifact.

7.4 Coupling Strengths and Horizons

The main coupling strengths are:

$$\alpha \in \{0.45, 0.50, 0.525\}. \quad (72)$$

The main horizons are:

$$T \in \{900, 1500, 2400\}. \quad (73)$$

Trajectories are divided into six segment boundaries for multi-step propagation analysis. Thus, rather than measuring only a start-to-end statistic, the probes ask whether viable macro-fiber structure persists through a sequence of segment transitions.

For example, a trajectory induces a macro-label sequence:

$$z_0 \rightarrow z_1 \rightarrow \cdots \rightarrow z_K, \quad (74)$$

where $K = 5$ for six segment boundaries and each z_i is determined by the chosen quotient.

7.5 Primary Quotient: Center of Mass

The primary quotient in the current witness is the center-of-mass quotient:

$$\kappa_{\text{COM}}(s_t^{AB}) = \text{bin} \left(\frac{x_t^A + x_t^B}{2} \right). \quad (75)$$

This quotient is deliberately simple. It compresses the joint state into a single collective coordinate and asks whether that coordinate supports certified viable transport through time.

The use of center of mass is not presented as universal. It is the current empirical witness quotient: the macro-map under which the toy substrate exposes the strongest audited propagation signature.

7.6 Control Quotients

The probes compare the COM quotient against several alternatives and controls.

- **Joint basin quotients** test basin-level structure and local transport alternatives.
- **Boundary or regime-sequence quotients** test pseudo-risk structure, including cases where apparent breadth or regime activity does not yield viable propagation.
- **Learned predictive k-means quotients** test whether a simple learned macro-map can recover the propagation structure without being handed COM bins as labels.
- **Identity-like diagnostics** test overfine descriptions.
- **All-one diagnostics** test collapsed descriptions.
- **Random or matched-label diagnostics** test whether label count alone explains the signal.

These controls matter because the quotient problem is load-bearing. A positive signal under COM is more meaningful when entropy-only, overfine, collapsed, random, and learned-fragmented alternatives fail in identifiable ways.

7.7 Null Baselines

Each quotient-indexed propagation statistic is compared against null baselines.

The **product baseline** estimates what propagation would look like if the two fields composed independently. It asks whether the coupled signal exceeds what is expected from separate single-field structure.

The **shuffled baseline** preserves marginal trajectory information while destroying coupling-specific joint correlations. It asks whether the joint signal depends on real coupled structure rather than marginal trajectory statistics.

The **independent calibration** uses $\alpha = 0$ or equivalent uncoupled conditions to check whether the propagation signal depends on attractive coupling.

A coupled witness is not established merely by having a positive propagation index. It should also separate from the relevant nulls:

$$\Delta_{\text{prod}}^{\kappa} > 0, \tag{76}$$

$$\Delta_{\text{shuf}}^{\kappa} > 0, \tag{77}$$

or otherwise produce an interpretable baseline-separated profile.

7.8 Certification Thresholds

The probes use operational thresholds to certify nodes, edges, and paths. The main thresholds are:

$$m_{\text{node}} = 0.005, \quad (78)$$

$$m_{\text{edge}} = 0.001, \quad (79)$$

$$\sigma_{\text{max}} = 0.65, \quad (80)$$

$$C_{\text{min}} = 0.70, \quad (81)$$

$$E_{\text{max}} = 0.20. \quad (82)$$

These correspond to minimum node fiber mass, minimum edge transport mass, maximum singleton fraction, minimum component preservation, and maximum lower-rank erasure.

The thresholds are not treated as universal constants. They are operational certification rules. Probe 12A therefore evaluates loose, main, and strict threshold variants to check whether the COM witness is threshold-stable.

7.9 Sampling and Uncertainty

The probes estimate propagation statistics using sampled trajectories, multiple random seeds, and bootstrap uncertainty estimates. The precise trajectory count, seed count, and bootstrap count vary by probe.

The current Probe 12 formal audit used a fallback scale:

$$N_{\text{traj}} = 3000, \quad N_{\text{seeds}} = 100, \quad N_{\text{bootstrap}} = 300. \quad (83)$$

The earlier robustness probes used larger or targeted runs where appropriate. The important methodological point is that the reported witness is not based on a single deterministic trajectory or a single seed. It is evaluated as a statistical propagation profile over sampled trajectories and null baselines.

7.10 Scope of the Substrate

The current substrate is intentionally narrow. It establishes only the following kind of claim:

In the F, T attractive two-field toy substrate, the COM quotient exposes a certified multi-step viable propagation witness.

It does not establish that COM is universal, that the same signature appears in higher-dimensional systems, that learned quotients can recover the witness, or that the result generalizes to real-world alignment, ecology, or sociology settings.

The substrate is useful precisely because it is small enough for failure modes to be visible. The next stage is not to overinterpret the substrate, but to use it to define the formal witness cleanly and then test whether nearby systems preserve, deform, or destroy the signature.

8 The COM Fiber-Transport Witness

This section defines the concrete toy-substrate witness currently supported by the experiments: certified multi-step viable propagation through center-of-mass fibers. The construction is not proposed as the final Omega object. It is a bounded quotient-relative witness under a specific macro-map.

The central object is:

$$\Omega_{\text{witness}} = \mathcal{P}_T^{\kappa_{\text{COM}}}(S_{\text{toy}}), \quad (84)$$

where S_{toy} is the toy multifield substrate, κ_{COM} is the center-of-mass quotient, and $\mathcal{P}_T^{\kappa_{\text{COM}}}$ denotes certified viable propagation through the induced macro-fibers over horizon T .

The important point is that this witness is not defined by entropy, raw survival, or raw reachability. It is a structured propagation object. Its scalar summary is useful for comparison, but the signature is the tuple of viability, certified transport, multi-step depth, component preservation, nonfragmentation, and null-baseline separation.

8.1 The Center-of-Mass Quotient

In the toy substrate, a joint state is:

$$s_t^{AB} = (s_t^A, s_t^B), \quad (85)$$

where each component state contains a one-dimensional position coordinate, x_t^A and x_t^B . The center-of-mass coordinate is:

$$c_t = \frac{x_t^A + x_t^B}{2}. \quad (86)$$

The COM quotient maps the joint state to a discrete macro-label:

$$z_t = \kappa_{\text{COM}}(s_t^{AB}) = \text{bin}(c_t), \quad (87)$$

where $\text{bin}(\cdot)$ is the fixed discretization used in the experiments.

This quotient is intentionally simple. It does not encode the full joint state, the full trajectory, the basin identity of each component, or the viability label directly. It asks whether the collective coordinate c_t supports a macro-fiber structure through which viable joint trajectories propagate.

The use of κ_{COM} should be understood as a witness choice, not as a claim of universality. The intended final theory would replace this hand-defined quotient with a minimal admissible predictive quotient κ^* , or with a stable family of admissible quotients $\mathcal{K}_{\text{adm}}$. At the current stage, κ_{COM} is the strongest empirical witness.

8.2 COM Fibers

Given a macro-label z at segment boundary t_i , the COM fiber is the set of viable joint trajectories whose COM macro-label equals z at that segment:

$$\mathcal{R}_{t_i}^{\text{COM}}(z) = \{\tau_{0:T}^{AB} \in \mathcal{F}_T^{AB} : \kappa_{\text{COM}}(s_{t_i}^{AB}) = z\}. \quad (88)$$

A fiber is therefore not merely a bin in COM-space. It is a set of viable micro-realizations supporting that macro-label. This distinction matters because a macro-state with no viable support, or only brittle one-off support, should not count as robust propagation structure.

For each fiber, the probes track quantities such as:

- fiber mass;
- fiber size;
- singleton status;

- small-fiber status;
- viable fiber fraction;
- component projection structure;
- component preservation;
- lower-rank erasure.

The fiber is the bridge between the macro-coordinate and the underlying viable trajectory set.

8.3 Certified Macro-Nodes

A COM macro-label z at segment t_i becomes a certified node only if its fiber has sufficient viable support and does not collapse into a degenerate micro-realization.

Let $M_{t_i}(z)$ denote the empirical viable mass of $\mathcal{R}_{t_i}^{\text{COM}}(z)$. Let $\sigma_{t_i}(z)$ denote a fragmentation diagnostic, and let $C_A(z)$ and $C_B(z)$ denote component preservation scores for the projections of the fiber onto fields A and B . Let $E_{\text{lower}}(z)$ denote lower-rank erasure.

A node is certified when:

$$M_{t_i}(z) \geq m_{\text{node}}, \quad (89)$$

$$\sigma_{t_i}(z) \leq \sigma_{\text{max}}, \quad (90)$$

$$C_A(z) \geq C_{\text{min}}, \quad (91)$$

$$C_B(z) \geq C_{\text{min}}, \quad (92)$$

$$E_{\text{lower}}(z) \leq E_{\text{max}}. \quad (93)$$

In the current implementation, the main thresholds are:

$$m_{\text{node}} = 0.005, \quad (94)$$

$$\sigma_{\text{max}} = 0.65, \quad (95)$$

$$C_{\text{min}} = 0.70, \quad (96)$$

$$E_{\text{max}} = 0.20. \quad (97)$$

These thresholds are operational certification rules, not derived constants of the theory. Probe 12A therefore evaluates loose, main, and strict threshold variants. The COM witness is stronger when the propagation readout is stable across these variants.

The singleton and small-fiber diagnostics are especially important. A quotient that produces many one-realization fibers is identity-like; it may appear informative while failing to provide robust macro-support. Certification therefore requires not only positive viability, but nondegenerate fiber support.

8.4 Certified Transport Edges

For adjacent segment boundaries t_i and t_{i+1} , a transport edge is an observed macro-transition:

$$z_i \rightarrow z_{i+1}, \quad (98)$$

where:

$$z_i = \kappa_{\text{COM}}(s_{t_i}^{AB}), \quad z_{i+1} = \kappa_{\text{COM}}(s_{t_{i+1}}^{AB}). \quad (99)$$

The transported viable mass of this edge is:

$$M_i(z_i, z_{i+1}) = \Pr \left(\tau_{0:T}^{AB} \in \mathcal{F}_T^{AB}, \kappa_{\text{COM}}(s_{t_i}^{AB}) = z_i, \kappa_{\text{COM}}(s_{t_{i+1}}^{AB}) = z_{i+1} \right), \quad (100)$$

estimated empirically from sampled trajectories.

An edge is certified when:

$$M_i(z_i, z_{i+1}) \geq m_{\text{edge}}, \quad (101)$$

$$z_i \text{ is a certified node}, \quad (102)$$

$$z_{i+1} \text{ is a certified node}. \quad (103)$$

The current main edge-mass threshold is:

$$m_{\text{edge}} = 0.001. \quad (104)$$

The certified edge condition ensures that propagation is not credited for transitions between unsupported or degenerate macro-labels. A macro-transition only counts when viable micro-realizations are transported from a certified source fiber into a certified target fiber.

This is the first point where the construction differs sharply from raw macro-entropy. A high number of possible macro-transitions is not enough. The transitions must carry viable fiber mass through certified support.

8.5 Certified Multi-Step Paths

A certified multi-step path is a sequence:

$$\gamma = z_0 \rightarrow z_1 \rightarrow \cdots \rightarrow z_K, \quad (105)$$

where each node is certified and each adjacent edge is certified.

Let Γ_T^{COM} denote the set of certified COM macro-paths over horizon T . For a path $\gamma \in \Gamma_T^{\text{COM}}$, define its path mass:

$$M(\gamma) = \Pr \left(\tau_{0:T}^{AB} \in \mathcal{F}_T^{AB} \text{ and follows macro-path } \gamma \right). \quad (106)$$

Define certified path survival to the final segment as the mass of viable trajectories that remain on certified paths through the final segment:

$$S_{\text{path}} = \sum_{\gamma \in \Gamma_T^{\text{COM}}} M(\gamma). \quad (107)$$

The multi-step nature of the path criterion is essential. Earlier transport signals can be local: a quotient may show one-step transport but fail to carry viable structure through the full segmented horizon. The COM witness requires multi-step certified propagation, not merely local transition density.

Thus, the witness is not:

there exists a viable transition,

but rather:

viable fiber-supported macro-structure is carried through a sequence of certified transitions over the horizon.

8.6 Component Preservation

Because the substrate is multifield, joint propagation is not sufficient by itself. A macro-channel can preserve a joint aggregate while erasing one of the components. Such a channel would be lower-rank collapse, not Omega-like propagation.

For each certified fiber $\mathcal{R}_{t_i}^{\text{COM}}(z)$, define component projections:

$$\pi_A(\mathcal{R}_{t_i}^{\text{COM}}(z)), \quad (108)$$

$$\pi_B(\mathcal{R}_{t_i}^{\text{COM}}(z)). \quad (109)$$

These projections track the induced lower-rank structure of fields A and B inside the joint fiber. Component preservation scores C_A and C_B measure whether the projected component structure remains non-erased along certified nodes and paths.

The requirement is:

$$C_A \geq C_{\min}, \quad C_B \geq C_{\min}, \quad (110)$$

with lower-rank erasure bounded by:

$$E_{\text{lower}} \leq E_{\max}. \quad (111)$$

This condition prevents a joint macro-coordinate from receiving credit merely because it preserves a coarse aggregate. The aggregate must be supported by component-realizing fibers.

8.7 The Viable Propagation Tuple

The COM signature is primarily a structured tuple, not a scalar. For a quotient κ , define:

$$\Omega_{\text{sig}_T}^{\kappa} = (p_{\text{viable}}, S_{\text{path}}, D_{\text{transport}}, C_A, C_B, E_{\text{lower}}, N_{\text{frag}}, \Delta_{\text{prod}}, \Delta_{\text{shuf}}). \quad (112)$$

The terms are:

- p_{viable} : viable trajectory mass;
- S_{path} : certified path survival to the final segment;
- $D_{\text{transport}}$: multi-step transport depth;
- C_A, C_B : component preservation scores;
- E_{lower} : lower-rank erasure;
- N_{frag} : nonfragmentation diagnostics, including singleton and small-fiber fractions;
- Δ_{prod} : propagation advantage over product baseline;
- Δ_{shuf} : propagation advantage over shuffled baseline.

The tuple is primary because different failure modes affect different coordinates. A quotient can have high breadth but poor propagation, high one-step transport but weak multi-step depth, high joint persistence but poor component preservation, or positive training prediction but severe fragmentation. Collapsing these into one number too early obscures the object.

8.8 Scalar Propagation Index

For compact reporting, the probes use a scalar viable propagation index. A representative form is:

$$\mathcal{P}_T^\kappa = S_{\text{path}} \cdot D_{\text{transport}} \cdot C_{\text{component}} \cdot N_{\text{nonfrag}}. \quad (113)$$

Here:

$$C_{\text{component}} = \min(C_A, C_B), \quad (114)$$

and N_{nonfrag} is a compact nonfragmentation factor derived from singleton and small-fiber diagnostics.

The exact scalar form is operational and may be revised. Its role is to support comparison across probes and baselines. It should not be mistaken for the full Omega object.

The scalar is valid only as a summary of a tuple that has already passed the following gates:

1. viability;
2. certified node and edge support;
3. multi-step propagation;
4. component preservation;
5. nonfragmentation;
6. positive separation from null baselines.

A high scalar-like value without these gates is not Omega-positive.

8.9 Null-Baseline Separation

The COM witness is evaluated against two primary null baselines.

The product baseline estimates what propagation would look like if the two fields composed independently. The shuffled baseline destroys coupling-specific joint correlations while preserving marginal trajectory information.

The corresponding deltas are:

$$\Delta_{\text{prod}}^\kappa = \mathcal{P}_{T,\text{coupled}}^\kappa - \mathcal{P}_{T,\text{product}}^\kappa, \quad (115)$$

$$\Delta_{\text{shuf}}^\kappa = \mathcal{P}_{T,\text{coupled}}^\kappa - \mathcal{P}_{T,\text{shuffled}}^\kappa. \quad (116)$$

The COM witness requires positive or interpretable separation from these nulls. A propagation structure that cannot separate from product or shuffled baselines is not evidence of a coupled multifield signature.

8.10 Why This Witness Is Not Entropy

The COM fiber-transport witness is not defined by maximizing entropy. Entropy appears only as a secondary breadth diagnostic. This distinction matters because bad quotients can launder entropy by overfragmenting the state space or by creating macro-labels that are numerous but not propagatively meaningful.

The relevant contrast is:

$$H(\text{macro futures}) \quad \text{versus} \quad \mathcal{P}_T^\kappa(\text{certified viable propagation}). \quad (117)$$

The former measures breadth. The latter measures whether viable structure is carried through certified fibers across time while preserving component support.

A quotient can be entropy-rich and propagation-poor. Such a case is classified as pseudo-risk or entropy-only structure, not as an Omega witness.

8.11 Interpretation of the COM Witness

The current interpretation is:

In the F, T attractive two-field toy substrate, the center-of-mass quotient exposes a macro-fiber channel through which viable joint structure propagates over multiple segment boundaries, with positive separation from product and shuffled baselines and with sufficient component preservation.

This is a bounded result. It does not imply that center-of-mass is the universal quotient, nor that κ_{COM} is the final κ^* . It shows that, in this substrate, there exists a concrete macro-coordinate under which the viable propagation signature becomes visible and survives controls.

The scientific role of κ_{COM} is therefore as a witness:

$$\exists \kappa \quad \text{such that} \quad \mathcal{P}_T^\kappa \text{ is positive, certified, component-preserving, and null-separated.} \quad (118)$$

The next formal objective is to characterize this witness precisely enough that it can be reproduced, criticized, and eventually generalized or replaced by a learned admissible quotient.

9 Empirical Witness from Probes 09–12

This section summarizes the empirical support for the current COM fiber-transport witness. The goal is not to present every intermediate probe artifact, but to identify the result that survived the validation sequence and the controls that constrained its interpretation.

The current witness is:

$$\Omega_{\text{witness}} = \mathcal{P}_T^{\kappa_{\text{COM}}}(S_{\text{toy}}), \quad (119)$$

where S_{toy} is the F, T attractive two-field toy substrate and κ_{COM} is the center-of-mass quotient.

9.1 The Three Probe Principles

The probe sequence was guided by three methodological principles:

1. **Minimal.** Each probe should test the smallest object capable of answering the current question. New substrates, new agents, and new theory layers are deferred until the current object is understood.
2. **Principled.** Each test should compare against relevant nulls, expose failure modes, and separate primary propagation metrics from secondary diagnostics such as entropy or breadth.
3. **Revelatory.** The probes should allow the object to show what it is, rather than forcing it to match an expected interpretation. A useful probe can confirm, weaken, redirect, or reject a candidate.

These principles shaped the transition from entropy-based single-field proxies to the current propagation-first multifield witness. They also constrained the learned-quotient work: learned κ methods were not allowed to count as success merely by increasing entropy, improving short-horizon prediction, or fragmenting the state space.

9.2 Probe 09: Robust Fiber Reachability

Probe 09 isolated the first clear multifield candidate: center-of-mass multi-step viable propagation through certified fibers.

The key question was whether the transport-like signal from earlier multifield probes represented genuine multi-step viable propagation or merely local one-step transport. Probe 09 introduced a fiber reachability graph over macro-labels and evaluated certified paths across multiple segment boundaries.

The main finding was that κ_{COM} produced positive viable propagation relative to null baselines across the tested F, T attractive corridor. Other candidate quotients, including basin-like alternatives and boundary-like controls, were informative but did not replace COM as the clean multi-step witness.

Probe 09 therefore shifted the status of COM from a promising coordinate to a candidate fiber-transport witness.

9.3 Probe 10: Perturbation Robustness

Probe 10 asked whether the COM witness was a one-off artifact of the exact toy substrate. It tested the F, T attractive corridor under perturbations to the potential shape, noise level, sink threshold, initial conditions, time discretization, and certification thresholds.

The main result was that COM viable propagation remained robust but not uniformly so. Failures were concentrated in harder noise and potential-shape perturbations, especially where component preservation weakened. This was interpreted as a meaningful domain boundary rather than estimator instability.

The important conclusion was:

COM viable propagation is not merely a one-off Probe 09 artifact. It survives substantial perturbation coverage in the toy substrate.

This result justified treating COM as a serious witness, while preserving the bounded interpretation that the witness remains substrate-local.

9.4 Probe 11: Learned Predictive Kappa

Probe 11 tested whether a simple learned quotient could recover the viable propagation structure without being handed COM bins as labels. The learned candidates were predictive k-means quotients trained to predict primitive future viable propagation targets under a simplicity penalty.

The result was mixed and ultimately negative for replacing COM. Some learned quotients found partial propagation-positive structure, especially at lower cluster counts, but the best validation models did not recover COM as a strong coordinate on heldout moderate perturbations. Higher- k models often improved validation prediction while fragmenting on test, losing component preservation, or becoming entropy-like.

This was an important negative result. It suggested that generic predictive clustering can see part of the signal, but does not yet learn the admissible macro-coordinate needed for robust viable propagation.

9.5 Probe 12: COM Audit and Learned-Kappa Diagnosis

Probe 12 separated two questions that had become entangled:

1. Is COM itself a coherent formal witness?
2. Why did learned predictive quotients fail to replace it?

Probe 12A audited the COM object directly. Probe 12B diagnosed learned-kappa failure modes. Probe 12C ran a small transition-aware learner smoke test.

The main Probe 12A summary is:

Metric	Value
Viable propagation index	0.2556
Δ vs product	+0.0678
Δ vs shuffled	+0.0673
Component-A preservation	1.0000
Component-B preservation	0.7893
Lower-rank erasure	0.1054
Singleton fraction	0.4567
Small-fiber fraction	0.7725
Certified node fraction	0.7841
Certified edge fraction	0.8739
Certified path survival	0.6853
Multi-step transport depth	0.6911

Table 1: Probe 12A COM witness summary under main thresholds.

The threshold sensitivity result was tight:

This threshold stability is important because it suggests that the COM witness is not a fragile artifact of one exact certification setting.

Probe 12B found that learned predictive k-means mostly failed by splitting COM fibers, merging incompatible COM fibers, and inflating small-fiber structure. High- k validation winners did not translate into heldout COM-like propagation. Lower- k learned quotients were more promising as partial quotients, but still did not replace COM.

Threshold Variant	Viable Propagation Index
Loose	0.2569
Main	0.2556
Strict	0.2537

Table 2: Probe 12A threshold sensitivity for the COM witness.

Probe 12C tested a transition-aware balanced learner. It improved the anatomy of the learned quotient at smoke-test scale, especially by reducing fragmentation, but did not yet justify scaling the learned-kappa route before formalizing COM.

9.6 Empirical Status of the Witness

The empirical status after Probes 09–12 is:

1. COM exposes positive certified multi-step viable propagation in the F, T attractive toy substrate.
2. The COM propagation signal separates from product and shuffled null baselines.
3. The witness survives perturbation testing, with meaningful degradation under harder noise and potential-shape variants.
4. The witness is stable under loose, main, and strict threshold variants in Probe 12A.
5. Learned predictive k-means quotients can find partial signal, but do not yet recover COM as a strong admissible quotient.
6. The current best interpretation is therefore a quotient-relative COM witness, not a universal or learned Omega object.

9.7 Interpretive Summary

The empirical sequence supports the following bounded claim:

In the F, T attractive two-field toy substrate, the center-of-mass quotient exposes certified multi-step viable propagation through macro-fibers. The witness is positive relative to null baselines, stable under threshold variation, and more coherent than the learned or pseudo-risk alternatives tested so far.

The same sequence also identifies the main bottleneck:

The quotient problem remains unresolved. COM is the current witness, not the final quotient theory.

Thus the probes do not prove universal Omega. They establish a concrete, audited toy signature of the kind of viable propagation the universal theory is meant to capture.

10 Controls and Rejected Alternatives

The COM witness is meaningful only because alternative explanations were tested and constrained. This section summarizes the main controls and rejected alternatives. The goal is not to dismiss these structures as useless. Several are informative diagnostics. The point is narrower: they do not currently replace COM as the formal toy witness.

10.1 Why Controls Matter

Omega-like propagation is vulnerable to several false positives:

- raw survival without recoverability;
- raw entropy without viable transport;
- one-step transport without multi-step propagation;
- macro-stability with component erasure;
- overfine quotients that fragment fibers;
- collapsed quotients that erase distinctions;
- learned quotients that predict local features but fail heldout propagation.

The controls are designed to detect these cases. A candidate quotient should not be credited merely because it produces high breadth, high path count, high short-horizon prediction, or a positive scalar under one threshold. It must support certified viable propagation through time while preserving component structure and separating from null baselines.

10.2 Boundary and Regime-Sequence Controls

Boundary-like or regime-sequence quotients are useful pseudo-risk controls. They can produce apparent richness, regime variation, or breadth-like structure. However, they may fail to support certified viable propagation.

The characteristic failure mode is:

$$\Delta H_{\text{breadth}} > 0 \quad \text{or} \quad \Delta R > 0, \quad \text{but} \quad \Delta \mathcal{P}_T^k \leq 0. \quad (120)$$

This is an entropy-only or breadth-only pseudo-risk pattern. It indicates that the quotient is detecting activity or distinction, but not Omega-relevant propagation.

Boundary controls are therefore important because they validate the demotion of entropy. They show that positive breadth does not imply viable propagation.

10.3 Joint Basin and Basin-Transition Quotients

Basin-like quotients are more serious alternatives. They can capture meaningful dynamical structure and, in earlier probes, sometimes produced strong local transport-like signals.

However, the current evidence does not make joint basin the primary witness. The basin quotients are informative because they expose basin-level structure, but they did not replace COM as the clean certified multi-step propagation coordinate in the final audit sequence.

The distinction is:

Basin quotients can detect local or structural transport; COM currently exposes the cleaner audited multi-step viable propagation witness.

This may change in future substrates. In the present toy system, basin quotients remain controls and comparators, not the formal witness.

10.4 Identity-Like Quotients

An identity-like quotient maps too much microscopic detail into distinct macro labels. This can artificially increase apparent richness:

$$H(\kappa_{\text{identity}}(s)) \uparrow . \quad (121)$$

But such a quotient often fails to abstract. Its fibers are small or singleton:

$$\text{singleton_fraction} \approx 1. \quad (122)$$

This is a failure because Omega propagation is not meant to reward microscopic relabeling. A macro-coordinate should expose supported structure. If almost every macro-label corresponds to a single realization, the quotient is not providing a robust fiber.

Identity-like quotients are therefore diagnostic of overfragmentation.

10.5 All-One and Collapsed Quotients

The opposite failure mode is a collapsed quotient. An all-one quotient maps all states or trajectories to the same macro-label:

$$\kappa_{\text{all}}(s) = z_0 \quad \text{for all } s. \quad (123)$$

This avoids fragmentation but destroys meaningful distinctions. It may produce large fibers, but those fibers are not informative about future propagation. The quotient cannot distinguish viable transport channels, traps, recoverability differences, or component-preserving versus component-erasing structure.

Collapsed quotients are therefore diagnostic of underfragmentation or excessive abstraction.

10.6 Random and Matched-Label Quotients

Random or matched-label quotients test whether label count alone can explain the observed signal. A random quotient may match a candidate's number of labels, fiber sizes, or marginal label distribution, but should not preserve coherent future viable propagation.

A learned or hand-defined quotient is more credible when it outperforms matched random controls on:

- heldout viable propagation;
- certified path survival;
- multi-step transport depth;
- component preservation;
- nonfragmentation.

Random controls therefore help distinguish real macro-structure from accidental label statistics.

10.7 Learned Predictive Kappa

Probe 11 tested whether simple learned predictive quotients could recover the COM witness. The learned quotients were trained to predict primitive future viable propagation targets under a simplicity penalty. This was a deliberately clean test: the learner was not trained to maximize the final Omega-positive metric.

The result was partial but insufficient. Lower- k learned quotients found some propagation-positive structure. Higher- k quotients often achieved better validation prediction, but failed on heldout moderate perturbations through fragmentation, component-preservation loss, or negative propagation deltas.

Probe 12 diagnosed the main learned-kappa failure modes:

- splitting COM fibers into too many learned labels;
- merging distinct COM fibers with incompatible propagation profiles;
- inflating small-fiber structure;
- winning validation prediction without heldout propagation;
- failing component-B preservation.

This result does not show that learned quotients are impossible. It shows that generic predictive k-means is not yet a replacement for COM.

10.8 Transition-Aware Learned Quotients

Probe 12C tested a transition-aware balanced learner at smoke-test scale. This was motivated by the hypothesis that simple k-means was too local: it predicted segment-level targets but did not adequately preserve trajectory-level fiber transport.

The transition-aware learner improved some anatomy diagnostics, especially fragmentation, but it did not yet justify scaling learned-kappa methods before formalizing COM. Its current status is:

promising as a future direction, not established as a replacement witness.

This motivates a later comparison between COM-supervised, COM-blind, and fiber-coherence learning methods.

10.9 Why COM Remains the Current Witness

COM remains the current witness because it satisfies the strongest combination of criteria in the current toy substrate:

1. positive certified multi-step viable propagation;
2. separation from product and shuffled null baselines;
3. perturbation robustness;
4. threshold stability;
5. component preservation;

6. lower-rank erasure control;
7. better coherence than learned or pseudo-risk alternatives.

This does not imply that COM is universal. It means that, among the tested quotients, COM is the clearest audited witness of the target structure.

10.10 Interpretive Role of Rejected Alternatives

The rejected alternatives are not failures of the research program. They define the boundaries of the object.

- Boundary controls show that entropy and breadth are not enough.
- Identity controls show that overfragmentation is not admissible.
- All-one controls show that collapsed abstraction is not admissible.
- Random controls show that label count is not enough.
- Learned-kappa failures show that predictive sufficiency and admissible fiber structure are not automatically recovered by generic clustering.
- Basin controls show that meaningful dynamical structure can exist without becoming the current formal witness.

Together, these controls make the COM result more credible. They show that the workflow is not simply selecting whichever quotient gives a positive number. It is rejecting plausible but weaker objects and isolating the one that best matches the viable-propagation signature in the current substrate.

11 What the Result Means

The current result should be interpreted as a bounded formal witness. It is more than a suggestive metaphor, but less than a general theory. The experiments support a quotient-relative toy signature of viable propagation:

$$\Omega_{\text{witness}} = \mathcal{P}_T^{\kappa_{\text{COM}}}(S_{\text{toy}}). \quad (124)$$

They do not yet support a universal, quotient-independent Omega object.

11.1 The Positive Claim

The positive claim is:

In the F, T attractive two-field toy substrate, the center-of-mass quotient exposes certified multi-step viable propagation through macro-fibers, with positive null-baseline separation, threshold stability, and component preservation.

This is the current Omega signature. It is not raw entropy, not survival alone, and not one-step transport. It is a structured propagation witness.

The witness has three important features.

First, it is **viability-gated**. Nonviable trajectories do not contribute to the propagation object except as failure modes.

Second, it is **multi-step**. The signal is not merely a local transition or short-horizon correlation. It requires certified propagation across segment boundaries.

Third, it is **fiber-supported**. The macro-labels are not treated as self-sufficient. They must be supported by viable micro-realizations with nondegenerate component projections.

11.2 Why This Is Nontrivial

The result is nontrivial because several weaker alternatives were available and did not survive the same constraints.

A raw entropy story would have predicted that the important object is breadth or the number of distinguishable futures. The probes instead show that entropy can be secondary or misleading. Breadth-like controls can produce apparent richness without certified propagation.

A raw survival story would have treated viable persistence as enough. The single-field and multifield probes instead distinguish persistence from recoverable, transportable viable structure.

A raw learned-quotient story would have expected generic predictive clustering to rediscover the relevant macro-map. Probe 11 and Probe 12 show that simple learned quotients find partial structure but often fragment, merge incompatible fibers, or fail heldout component preservation.

The COM witness therefore occupies a specific position: it is not merely the largest entropy score, not merely a surviving trajectory class, and not merely what an unsupervised learner happens to find. It is the quotient under which the current toy substrate most clearly exposes certified viable propagation.

11.3 The Role of COM

COM should be interpreted as a witness coordinate. It is not claimed to be universal. Its role is to demonstrate existence in the current substrate:

$$\exists \kappa \text{ such that } \mathcal{P}_T^\kappa \text{ is positive, certified, component-preserving, and null-separated.} \quad (125)$$

In this toy substrate, the witness coordinate is κ_{COM} . Future work may show that COM is a special case of a broader quotient family, that it deforms smoothly across nearby substrates, or that it should be replaced by a learned admissible quotient. Those possibilities remain open.

The current evidence supports the more modest statement:

$$\mathcal{P}_T^{\kappa_{\text{COM}}}(S_{\text{toy}}) > \mathcal{P}_{T,\text{null}}^{\kappa_{\text{COM}}}(S_{\text{toy}}) \quad (126)$$

under the tested nulls and certification rules.

11.4 The Quotient Problem Remains Central

The result also clarifies the main theoretical bottleneck. Omega is intended to be universal or quotient-stable, but the current witness is quotient-relative.

If a minimal predictive quotient κ^* exists, then the long-term object may be written:

$$\Omega_T(S) = \mathcal{P}_T^{\kappa^*}(S). \quad (127)$$

If no such quotient is unique or stable, the appropriate object may instead be a family:

$$\{\mathcal{P}_T^{\kappa}(S) : \kappa \in \mathcal{K}_{\text{adm}}\}. \quad (128)$$

The COM witness does not decide between these possibilities. It makes the question concrete. It shows one admissible-looking quotient in one toy substrate and gives future work a target: characterize when such quotients exist, how stable they are, and whether they can be learned or derived.

11.5 What This Does Not Show

The current result does not show that Omega is proven. It does not show that COM is universal. It does not show that $\kappa_{\text{COM}} = \kappa^*$. It does not show that the same signature appears in richer substrates, higher dimensions, many-agent systems, or real-world settings.

It also does not show direct implications for alignment, ecology, sociology, or value theory. Those domains motivate the project, but the evidential bridge has not yet been built. At present, the result is a toy-substrate formal witness, not an applied theory.

The result should therefore not be stated as:

Omega has been found.

It should be stated as:

A quotient-relative toy witness of Omega-like viable propagation has been isolated and audited.

11.6 Why the Result Still Matters

The result matters because it turns an ambitious theoretical foreground into a specific mathematical object with tests and failure modes. It provides:

- a concrete quotient;
- a viable trajectory set;
- a fiber construction;
- certified node, edge, and path conditions;
- component-preservation diagnostics;
- null baselines;
- threshold sensitivity checks;
- rejected alternatives.

This is enough to define a research program around the object. The next questions are not purely philosophical. They are operational:

1. Can the COM witness be formalized cleanly enough for reproduction?

2. Does it survive nearby substrate generalization?
3. Can learned quotients recover it or an equivalent admissible structure?
4. Does the witness belong to a stable quotient family?
5. Can the certification thresholds be derived, reduced, or replaced by a cleaner invariant?

The current result matters because these questions are now precise enough to test.

11.7 Summary Interpretation

The best current interpretation is:

The Omega program aims at a universal formal theory of viable propagation. The current work has not reached that target. It has isolated a quotient-relative toy witness: certified multi-step viable propagation through COM fibers in a two-field substrate. This witness is nontrivial, controlled, and bounded. Its main theoretical bottleneck is the quotient problem.

This is the strongest honest claim supported by the present evidence.

12 Open Problems

The current artifact identifies a quotient-relative toy witness. It does not complete the Omega program. This section collects the open problems exposed by the probe sequence and by the COM witness formalization.

The most important open problem is the quotient problem. The other open problems are downstream of it: generalization, formal invariance, learned quotient recovery, and interpretation across domains.

12.1 The Quotient Problem

The current witness is defined relative to the center-of-mass quotient:

$$\Omega_{\text{witness}} = \mathcal{P}_T^{\kappa^{\text{COM}}}(S_{\text{toy}}). \quad (129)$$

The intended final object is more ambitious:

$$\Omega_T(S) = \mathcal{P}_T^{\kappa^*}(S), \quad (130)$$

where κ^* is a minimal admissible predictive quotient. The main open question is whether such a quotient exists, whether it is unique, and whether it can be characterized without hand-selecting a coordinate such as COM.

There are several possibilities:

1. **Unique quotient.** A natural κ^* exists and is recoverable from the system's viable propagation structure.
2. **Equivalence class.** There is no unique quotient, but there is a stable equivalence class of admissible quotients that expose the same propagation structure.

3. **Admissible family.** Omega is best represented as a family of quotient-indexed propagation signatures:

$$\{\mathcal{P}_T^\kappa(S) : \kappa \in \mathcal{K}_{\text{adm}}\}. \quad (131)$$

4. **Description-relative object.** Omega is fundamentally a property of a system together with a description:

$$(S, \kappa) \mapsto \mathcal{P}_T^\kappa(S). \quad (132)$$

The current evidence supports the last form most directly and leaves the others open. The COM witness shows that one useful quotient exists in the toy substrate. It does not yet show that the quotient is unique, universal, or learnable.

12.2 Admissibility Criteria

If Omega depends on a quotient, then the theory requires criteria for admissibility. An arbitrary quotient cannot be allowed, because arbitrary quotients can manufacture false positives.

An admissible quotient should avoid at least four failure modes:

1. **Collapse:** all relevant states are mapped to one label, erasing distinctions needed for propagation.
2. **Overfragmentation:** microscopic details are split into many labels, producing identity-like fibers.
3. **Entropy laundering:** apparent richness is created by label choice rather than viable propagation.
4. **Component erasure:** a joint macro-structure is preserved by destroying lower-rank component support.

The current operational admissibility checks include fiber mass, singleton fraction, small-fiber fraction, label imbalance, component preservation, lower-rank erasure, threshold stability, and null-baseline separation. These are useful diagnostics, but they are not yet a complete theory of admissibility.

A future formal theory should explain why these diagnostics, or cleaner successors to them, are the right constraints.

12.3 The Learned-Quotient Problem

Probes 11–12 show that simple learned predictive quotients can find partial signal but do not recover the COM witness as a strong coordinate. This exposes a deeper learning problem.

A learned quotient should not merely predict local future labels. It should preserve fiber transport structure across time. The failure modes of simple predictive k-means suggest that learned quotients need to account for:

- sequence-level propagation;
- fiber coherence;
- component preservation;
- heldout perturbation robustness;

- avoidance of fiber splitting and incompatible merging;
- long-horizon prediction rather than only local predictability.

This leads to a sharper learning objective. Instead of learning labels that predict local targets, the learner should approximate an admissible quotient for viable propagation:

$$\kappa^* = \arg \min_{\kappa} [\mathcal{L}_{\text{prop}}(\kappa) + \lambda \mathcal{L}_{\text{adm}}(\kappa)], \quad (133)$$

where $\mathcal{L}_{\text{adm}}$ penalizes fragmentation, component erasure, and instability under perturbation.

However, if the learner is trained to reproduce COM directly, then success no longer shows that the quotient emerged from the dynamics alone. A useful future program may therefore require two branches:

1. a COM-supervised or COM-curriculum branch, useful as an engineering bridge;
2. a COM-blind fiber-coherence branch, useful as the cleaner scientific test.

The learned-quotient problem remains unresolved.

12.4 The Generalization Problem

The COM witness is currently established only in the F, T attractive two-field toy substrate. The next major empirical question is whether the signature survives nearby changes.

Generalization should be tested in stages:

1. nearby perturbations of the same substrate;
2. nearby initial condition corridors;
3. other coupling strengths and coupling types;
4. alternate toy potentials or viability boundaries;
5. higher-dimensional fields;
6. more than two coupled fields;
7. richer agent-like or ecological substrates.

The key question is not merely whether a positive number appears elsewhere. The key question is whether the same kind of certified viable propagation survives:

$$\mathcal{P}_T^\kappa(S') > \mathcal{P}_{T, \text{null}}^\kappa(S') \quad (134)$$

with component preservation, nonfragmentation, and threshold stability.

Generalization also intersects with the quotient problem. A nearby substrate may preserve the COM witness, require a smoothly deformed COM-like quotient, or require an entirely different admissible quotient. These are different outcomes with different theoretical meanings.

12.5 The Threshold Problem

The current certification rules use operational thresholds:

$$m_{\text{node}} = 0.005, \quad (135)$$

$$m_{\text{edge}} = 0.001, \quad (136)$$

$$\sigma_{\text{max}} = 0.65, \quad (137)$$

$$C_{\text{min}} = 0.70, \quad (138)$$

$$E_{\text{max}} = 0.20. \quad (139)$$

Probe 12A suggests that the COM witness is stable under loose, main, and strict threshold variants. That is encouraging, but it does not solve the threshold problem.

A mature theory should explain whether these thresholds are:

- merely practical finite-sample certification choices;
- approximations to scale-dependent admissibility constraints;
- replaceable by threshold-free quantities;
- derivable from statistical confidence or information criteria.

The present artifact treats thresholds as operational gates. It does not claim that they are fundamental constants.

12.6 The Scalarization Problem

The scalar viable propagation index is useful for reporting:

$$\mathcal{P}_T^\kappa = S_{\text{path}} \cdot D_{\text{transport}} \cdot C_{\text{component}} \cdot N_{\text{nonfrag}}. \quad (140)$$

But the scalar is not the full object. The full signature is structured:

$$\Omega_{\text{sig}_T}^\kappa = (p_{\text{viable}}, S_{\text{path}}, D_{\text{transport}}, C_A, C_B, E_{\text{lower}}, N_{\text{frag}}, \Delta_{\text{prod}}, \Delta_{\text{shuf}}). \quad (141)$$

The scalarization problem asks whether there is a principled way to combine these coordinates, or whether Omega should remain a partially ordered profile.

A mature formulation may need to preserve the profile form, because different failure modes are visible in different coordinates. A single scalar may be too compressive.

12.7 The Component-Preservation Problem

In multifield systems, a joint macro-state can persist while erasing one of the components. The current probes treat this as lower-rank erasure and require component preservation.

The open question is how this should generalize. In richer systems, components may be nested, overlapping, or observer-dependent. It may not be obvious what counts as a lower-rank field. The theory therefore needs a general account of component structure.

Possible directions include:

- explicitly given component projections;

- learned or inferred component decompositions;
- hierarchical fiber structures;
- category-like or sheaf-like descriptions of local-to-global support;
- scale-indexed preservation criteria.

The current two-field substrate avoids this complexity by giving the component split explicitly. General systems will not always do so.

12.8 The Interpretation Problem

The long-term motivation of Omega includes alignment, ecology, sociology, and value theory. The current result does not yet license direct claims in those domains. It does, however, clarify the kind of structure such applications would need to identify: viable propagation through admissible macro-structure with component preservation.

The interpretation problem asks how to translate the formal primitives into domain-specific objects:

- What is viability in an institution, ecosystem, or AI-human system?
- What are the components?
- What counts as lower-rank erasure?
- What is the admissible quotient?
- What are the relevant null baselines?
- What perturbations should robustness be tested against?

Until these questions are answered, applied interpretations remain speculative.

12.9 Summary of Open Problems

The current project has moved from an abstract ambition to a concrete witness. The remaining open problems are now sharper:

1. characterize admissible quotients;
2. determine whether κ^* exists;
3. distinguish quotient-relative from quotient-stable Omega;
4. improve learned quotient recovery;
5. test nearby substrate generalization;
6. replace or justify operational thresholds;
7. preserve structured profiles rather than over-scalarizing;
8. generalize component preservation beyond explicitly given two-field systems.

These open problems do not weaken the current witness. They define the path from a quotient-relative toy signature toward the intended universal Omega object.

13 Future Directions

The current work has isolated and audited a quotient-relative COM witness in a toy multifold substrate. The next steps should preserve the same methodological discipline used in the probe sequence: minimal, principled, and revelatory. The goal is not to broaden the theory as quickly as possible, but to determine what the current witness is, what it depends on, and whether it can be generalized.

13.1 Immediate Next Step: Formal COM Fiber Transport

The immediate next step is to formalize the COM witness as a clean object:

$$\Omega_{\text{witness}} = \mathcal{P}_T^{\kappa_{\text{COM}}}(S_{\text{toy}}). \quad (142)$$

This should include:

- a canonical definition of COM fibers;
- node, edge, and path certification rules;
- the structured viable-propagation tuple;
- the scalar propagation index as a secondary summary;
- null-baseline definitions;
- threshold sensitivity;
- component-preservation diagnostics;
- known failure modes.

The purpose of this step is reproducibility. A reader should be able to identify what the witness is, how it is computed, and what would count as failure.

13.2 Canonical Reproduction Table

After the formal definition, the next deliverable should be a small canonical reproduction table. The table should focus only on the current witness corridor:

$$(F, T), \quad \text{attractive coupling}, \quad \alpha \in \{0.45, 0.50, 0.525\}, \quad T \in \{900, 1500, 2400\}. \quad (143)$$

It should report, for κ_{COM} :

- viable propagation index;
- certified path survival;
- multi-step transport depth;
- component preservation;
- lower-rank erasure;

- singleton and small-fiber fractions;
- Δ_{prod} ;
- Δ_{shuf} ;
- bootstrap intervals.

This table should become the reference artifact for the toy signature.

13.3 Quotient Stability Before Broad Generalization

Generalization should begin with quotient stability, not with large new substrates. The first question is whether the COM witness survives nearby changes in a recognizable form.

There are three possible outcomes:

1. **Same quotient stability.** The same COM quotient exposes viable propagation in nearby substrates.
2. **Smooth quotient deformation.** A COM-like quotient, adjusted smoothly to the substrate, exposes the same propagation structure.
3. **Quotient replacement.** COM fails, but another admissible quotient exposes a related propagation structure.

These outcomes have different theoretical meanings. Same-quotient stability would strengthen the case for COM-like structure. Smooth deformation would suggest a quotient family. Quotient replacement would push the theory toward a more general admissibility framework.

13.4 Horizon and Feature Ablations for Learned Quotients

Before scaling learned κ methods, the project should diagnose why simple predictive k-means fails. A useful next learned-quotient probe should focus on horizon and feature ablations.

Key questions include:

- Do learned quotients perform well at short horizons but fail at long horizons?
- Does COM become more important as the horizon becomes propagation- relevant?
- Does adding COM-like features help, or merely teach the answer?
- Does removing COM-like features destroy the partial learned signal?
- Does a fiber-coherence penalty reduce splitting and merging?
- Does transition-aware training improve heldout propagation rather than only validation prediction?

The goal is not to produce a powerful learner immediately. The goal is to learn which aspects of the quotient problem are blocking recovery.

13.5 COM-Supervised Versus COM-Blind Learning

Future learned-quotient work should separate two branches.

The first is a COM-supervised or COM-curriculum branch. In this branch, COM is used as a teaching signal, auxiliary target, or curriculum. This can be useful engineering: it may produce learners that approximate the known witness and help identify which features are needed.

The second is a COM-blind fiber-coherence branch. In this branch, the learner is not trained to reproduce COM. It is trained to preserve viable propagation, fiber coherence, component support, and heldout robustness.

Both branches are useful, but they answer different questions. COM-supervised success shows that the witness can be approximated. COM-blind success would be stronger evidence that an admissible quotient can emerge from the dynamics.

13.6 Nearby Substrate Generalization

Once the COM witness is formalized, the next empirical generalization step should be modest. The project should test nearby substrates before moving to richer systems.

Candidate tests include:

- nearby initial corridors;
- small changes to potential geometry;
- alternative viability boundaries;
- attractive coupling variants;
- weak repulsive or mixed coupling controls;
- slightly higher-dimensional toy fields.

The criterion should remain propagation-first. A generalization result should not be accepted merely because entropy increases or a scalar improves. It should show certified viable propagation, component preservation, nonfragmentation, and null-baseline separation.

13.7 Toward an Admissible Quotient Family

If the COM witness survives only in some settings, the next theoretical object may be an admissible quotient family:

$$\mathcal{K}_{\text{adm}} = \{\kappa : \kappa \text{ preserves viable propagation under admissibility constraints}\}. \quad (144)$$

The corresponding Omega object would be family-valued:

$$\Omega_T(S) = \{\mathcal{P}_T^\kappa(S) : \kappa \in \mathcal{K}_{\text{adm}}\}. \quad (145)$$

This may be more realistic than a unique quotient. The goal would then be to characterize the admissible family, identify equivalence classes within it, and understand which signatures are quotient-stable.

13.8 Toward the Universal Object

The universal Omega object remains the long-term aim. The path from the current witness to that object likely requires four milestones:

1. **Witness formalization.** Define the current COM witness cleanly.
2. **Quotient stability.** Show that the witness survives nearby substrate variation under the same or smoothly related quotients.
3. **Admissible quotient theory.** Characterize the family of quotients under which viable propagation is meaningful.
4. **Generalized propagation signature.** Define Omega as either $\mathcal{P}_T^{\kappa^*}$ or as a quotient-family object over $\mathcal{K}_{\text{adm}}$.

Only after these milestones should the project make strong claims about alignment, ecology, sociology, or value theory.

13.9 Methodological Guardrails

Future work should preserve the three probe principles.

- **Minimal:** add the fewest new moving parts needed to answer the current question.
- **Principled:** define nulls, controls, diagnostics, and failure modes before interpreting results.
- **Revelatory:** allow the object to fail, deform, or reveal that the current interpretation is wrong.

These guardrails are part of the result. They are what allowed the project to demote entropy, reject learned-fragmented quotients, and isolate the COM witness without overclaiming it.

13.10 Summary Roadmap

The near-term roadmap is:

1. formalize the COM fiber-transport witness;
2. produce a canonical reproduction table;
3. run horizon and feature ablations for learned quotients;
4. compare COM-supervised and COM-blind fiber-coherence learning;
5. test nearby substrate generalization;
6. characterize admissible quotient families;
7. return to the universal Omega object with stronger evidence.

This sequence preserves the ambition of the theory while keeping the evidential claims bounded. The goal is universal Omega, but the path begins with a precise toy witness.

14 Limitations

This artifact reports a bounded toy-substrate witness. Its purpose is to define the current Omega-like signature precisely enough to be reproduced, criticized, and generalized. It does not claim that the broader Omega program is complete.

14.1 Toy-Substrate Limitation

The current witness is established only in a deliberately small toy multifold substrate. The system consists of two coupled one-dimensional stochastic fields, a viability constraint, and an F, T attractive corridor. This is a controlled validation environment, not a realistic model of natural, social, ecological, or AI systems.

The result therefore supports:

a toy-substrate COM fiber-transport witness,

not:

a general law of viable propagation across all systems.

The toy substrate is valuable because it makes failure modes visible. But the same simplicity limits how far the result can be interpreted.

14.2 Quotient-Relative Limitation

The current witness is quotient-relative. It is defined under κ_{COM} :

$$\Omega_{\text{witness}} = \mathcal{P}_T^{\kappa_{\text{COM}}}(S_{\text{toy}}). \quad (146)$$

It does not yet establish the intended final object:

$$\Omega_T(S) = \mathcal{P}_T^{\kappa^*}(S). \quad (147)$$

This is the central limitation. COM is a hand-defined coordinate. It has been audited and tested against controls, but it has not been derived as the unique or minimal admissible quotient. Probes 11–12 show that simple learned quotients do not yet recover it.

The result is therefore best interpreted as:

there exists at least one useful quotient exposing viable propagation in this toy substrate,

not:

the quotient problem is solved.

14.3 Threshold Limitation

The witness depends on operational certification thresholds, including minimum node mass, minimum edge mass, singleton fraction, component preservation, and lower-rank erasure bounds.

Probe 12A shows that the COM witness is stable across loose, main, and strict threshold variants. This is encouraging, but it does not make the thresholds fundamental. At present, the thresholds are finite-sample certification rules.

A mature theory should either derive these thresholds, replace them with confidence-based or scale-dependent criteria, or define a threshold-free functional.

14.4 Scalarization Limitation

The scalar viable propagation index is a useful reporting device, but it is not the full object. The full signature is a structured profile:

$$\Omega_{\text{sig}_T}^{\kappa} = (p_{\text{viable}}, S_{\text{path}}, D_{\text{transport}}, C_A, C_B, E_{\text{lower}}, N_{\text{frag}}, \Delta_{\text{prod}}, \Delta_{\text{shuf}}). \quad (148)$$

A single scalar can obscure failure modes. For example, a quotient might show positive propagation but fail component preservation, or show high breadth but fail multi-step depth. The scalar should therefore be treated as a compact summary of a profile that must be inspected.

14.5 Learned-Quotient Limitation

The current work does not solve learned quotient recovery. Probe 11 found partial learned signal, and Probe 12 diagnosed why simple predictive k-means fails, but no learned quotient currently replaces COM.

This limits the generality of the witness. If a future theory requires κ^* to be learned from dynamics alone, then the current evidence is incomplete. The learned route remains important, but it is not yet the main source of support for the COM witness.

14.6 Component-Structure Limitation

The toy substrate has an explicit two-field component structure:

$$S^{AB} = S^A \times S^B. \quad (149)$$

This makes component preservation and lower-rank erasure relatively easy to define. In richer systems, components may be nested, overlapping, learned, socially defined, or observer-dependent. The present work does not solve that general component-identification problem.

Thus, component preservation in the current artifact should be understood as a controlled toy version of a harder general requirement.

14.7 Generalization Limitation

The current result has not yet been shown to generalize to:

- other initial corridors;
- other coupling families;
- other toy potentials;
- higher-dimensional fields;
- many-field systems;
- agentic systems;
- ecological, social, or alignment-relevant systems.

The future roadmap includes nearby generalization, but this artifact does not claim it.

14.8 Interpretive Limitation

The long-term motivation of Omega includes alignment, ecology, sociology, and value theory. The current result does not yet justify direct claims in those domains.

At most, the current work provides a candidate formal pattern:

viable propagation through admissible macro-structure with component preservation.

Whether that pattern maps to real-world systems remains an open empirical and conceptual question.

14.9 Normative Limitation

Omega is motivated by viability and propagation, but this artifact does not derive a complete normative theory. It does not show that viable propagation is the only value, the highest value, or a moral axiom. It shows only that viable propagation can be formalized and witnessed in a toy setting.

A system might value termination, sacrifice, transformation, or goals not captured by persistence of future-bearing structure. Such systems are not necessarily irrational; they may simply fall outside the scope of the current Omega framing. This limitation should be kept explicit in future applications.

14.10 Summary of Limitations

The strongest honest statement is:

The current work identifies a quotient-relative COM fiber-transport witness in a toy multifield substrate. It does not establish universal Omega, solve the quotient problem, derive the thresholds, recover the witness with learned quotients, or justify direct applications to real-world domains.

These limitations do not invalidate the witness. They define what remains to be done.

A Operational Definitions

This appendix collects operational definitions used in the current toy-substrate probes. These definitions are implementation-facing. They are included to make the witness reproducible and to clarify which quantities are formal primitives, which are diagnostics, and which are finite-sample certification rules.

A.1 Viable Trajectory Mass

Let $\mathcal{T}_T$ be the sampled trajectory set over horizon T , and let $\mathcal{F}_T \subseteq \mathcal{T}_T$ be the viable subset. The empirical viable trajectory mass is:

$$p_{\text{viable}} = \frac{|\mathcal{F}_T|}{|\mathcal{T}_T|}. \tag{150}$$

In the multifield setting, $\mathcal{F}_T$ is replaced by $\mathcal{F}_T^{AB}$.

A.2 Fiber Size and Fiber Mass

For a quotient κ , macro-label z , and segment boundary t_i , the empirical fiber is:

$$\mathcal{R}_{t_i}^\kappa(z) = \{\tau_{0:T} \in \mathcal{F}_T : \kappa(s_{t_i}) = z\}. \quad (151)$$

The fiber size is:

$$n_{t_i}^\kappa(z) = |\mathcal{R}_{t_i}^\kappa(z)|. \quad (152)$$

The empirical fiber mass is:

$$M_{t_i}^\kappa(z) = \frac{n_{t_i}^\kappa(z)}{|\mathcal{F}_T|}. \quad (153)$$

A.3 Singleton Fraction

A singleton fiber is a fiber containing exactly one sampled viable realization. The singleton fraction is:

$$\text{singleton_fraction} = \frac{\#\{z : n^\kappa(z) = 1\}}{\#\{z : n^\kappa(z) > 0\}}. \quad (154)$$

When computed by segment, use $n_{t_i}^\kappa(z)$. When reported as a global diagnostic, aggregate over segment labels as specified in the probe output.

Singleton fraction diagnoses identity-like overfragmentation.

A.4 Small-Fiber Fraction

Given a fixed threshold m_{small} , the small-fiber fraction is:

$$\text{small_fiber_fraction} = \frac{\#\{z : 0 < n^\kappa(z) < m_{\text{small}}\}}{\#\{z : n^\kappa(z) > 0\}}. \quad (155)$$

In the current probes:

$$m_{\text{small}} = 5. \quad (156)$$

Small-fiber fraction is a softer version of singleton fraction. It detects near-identity fragmentation even when fibers are not literal singletons.

A.5 Label Imbalance

Let p_z be the empirical frequency of macro-label z among nonempty labels:

$$p_z = \frac{n^\kappa(z)}{\sum_{z'} n^\kappa(z')}. \quad (157)$$

The label entropy is:

$$H_Z = - \sum_z p_z \log p_z. \quad (158)$$

For k nonempty labels, label imbalance is:

$$\text{label_imbalance} = 1 - \frac{H_Z}{\log k}. \quad (159)$$

A value near 0 indicates a balanced label distribution. A value near 1 indicates strong imbalance.

A.6 Certified Node

A node (z, t_i) is certified if it passes the operational gates:

$$M_{t_i}^\kappa(z) \geq m_{\text{node}}, \quad (160)$$

$$\text{frag}_{t_i}^\kappa(z) \leq \sigma_{\max}, \quad (161)$$

$$C_A(z, t_i) \geq C_{\min}, \quad (162)$$

$$C_B(z, t_i) \geq C_{\min}, \quad (163)$$

$$E_{\text{lower}}(z, t_i) \leq E_{\max}. \quad (164)$$

Here $\text{frag}_{t_i}^\kappa(z)$ denotes the relevant fragmentation diagnostic for that node or its aggregate reporting context.

The current main thresholds are:

$$m_{\text{node}} = 0.005, \quad (165)$$

$$\sigma_{\max} = 0.65, \quad (166)$$

$$C_{\min} = 0.70, \quad (167)$$

$$E_{\max} = 0.20. \quad (168)$$

A.7 Transport Edge Mass

For adjacent segment boundaries t_i and t_{i+1} , the empirical transported mass from z_i to z_{i+1} is:

$$M_i^\kappa(z_i, z_{i+1}) = \frac{|\{\tau_{0:T} \in \mathcal{F}_T : \kappa(s_{t_i}) = z_i, \kappa(s_{t_{i+1}}) = z_{i+1}\}|}{|\mathcal{F}_T|}. \quad (169)$$

A.8 Certified Edge

An edge (z_i, z_{i+1}, t_i) is certified if:

$$M_i^\kappa(z_i, z_{i+1}) \geq m_{\text{edge}}, \quad (170)$$

$$\text{CertNode}(z_i, t_i; \kappa) = 1, \quad (171)$$

$$\text{CertNode}(z_{i+1}, t_{i+1}; \kappa) = 1. \quad (172)$$

The current main edge threshold is:

$$m_{\text{edge}} = 0.001. \quad (173)$$

A.9 Certified Path

A certified path is a macro-label sequence:

$$\gamma = (z_0, z_1, \dots, z_K) \quad (174)$$

such that each node and edge in the sequence is certified:

$$\text{CertNode}(z_i, t_i; \kappa) = 1 \quad \forall i, \quad (175)$$

$$\text{CertEdge}(z_i, z_{i+1}, t_i; \kappa) = 1 \quad \forall i < K. \quad (176)$$

The certified path mass is:

$$M^\kappa(\gamma) = \frac{|\{\tau_{0:T} \in \mathcal{F}_T : \kappa(s_{t_i}) = z_i \text{ for all } i\}|}{|\mathcal{F}_T|}. \quad (177)$$

A.10 Certified Path Survival to Final Segment

The certified path survival to final segment is:

$$S_{\text{path}}^\kappa = \sum_{\gamma \in \Gamma_T^\kappa} M^\kappa(\gamma), \quad (178)$$

where Γ_T^κ is the set of certified paths over horizon T .

This measures the viable trajectory mass that remains supported by certified macro-fiber paths through the full segmented horizon.

A.11 Multi-Step Transport Depth

The multi-step transport depth, $D_{\text{transport}}^\kappa$, summarizes how far certified transport persists across segment boundaries.

A generic depth-weighted form is:

$$D_{\text{transport}}^\kappa = \sum_{j=1}^K w_j \Pr(\text{certified path survives through segment } j), \quad (179)$$

with weights such as:

$$w_j = \frac{j}{K}. \quad (180)$$

The exact implementation should be specified in the code-facing reproduction artifact. Conceptually, this quantity distinguishes one-step transport from multi-step viable propagation.

A.12 Component Preservation

For a joint fiber $\mathcal{R}_{t_i}^{\kappa, AB}(z)$, component projections are:

$$\pi_A \left(\mathcal{R}_{t_i}^{\kappa, AB}(z) \right), \quad (181)$$

$$\pi_B \left(\mathcal{R}_{t_i}^{\kappa, AB}(z) \right). \quad (182)$$

Component preservation scores C_A and C_B measure whether these projected component structures remain supported along certified nodes or paths.

The current implementation reports these as normalized diagnostics in $[0, 1]$, with larger values indicating stronger preservation. The main certification threshold is:

$$C_A, C_B \geq 0.70. \quad (183)$$

The exact implementation should be linked to the canonical reproduction script.

A.13 Lower-Rank Erasure

Lower-rank erasure measures collapse of component support inside a joint propagation channel. It is reported as a normalized diagnostic:

$$E_{\text{lower}} \in [0, 1], \quad (184)$$

where larger values indicate more erasure.

The current main certification threshold is:

$$E_{\text{lower}} \leq 0.20. \quad (185)$$

This diagnostic prevents a joint macro-channel from receiving credit if it preserves an aggregate while eliminating the viability of one component.

A.14 Viable Propagation Index

The scalar viable propagation index is a compact summary of the structured signature. A representative form is:

$$\mathcal{P}_T^\kappa = S_{\text{path}}^\kappa \cdot D_{\text{transport}}^\kappa \cdot C_{\text{component}}^\kappa \cdot N_{\text{nonfrag}}^\kappa. \quad (186)$$

The component factor is:

$$C_{\text{component}}^\kappa = \min(C_A, C_B). \quad (187)$$

The nonfragmentation factor is derived from singleton and small-fiber diagnostics. The scalar index is not the full object; it is a reporting summary of a structured profile.

A.15 Propagation Deltas

For any quotient-indexed propagation statistic $\mathcal{P}_T^\kappa$, the product-baseline delta is:

$$\Delta_{\text{prod}}^\kappa = \mathcal{P}_{T,\text{coupled}}^\kappa - \mathcal{P}_{T,\text{product}}^\kappa. \quad (188)$$

The shuffled-baseline delta is:

$$\Delta_{\text{shuf}}^\kappa = \mathcal{P}_{T,\text{coupled}}^\kappa - \mathcal{P}_{T,\text{shuffled}}^\kappa. \quad (189)$$

Positive deltas indicate that the coupled system exceeds the corresponding null baseline under the same quotient and certification scheme.

A.16 Threshold Variants

The current probes use three threshold variants:

Variant	Node Mass	Edge Mass
Loose	0.0025	0.0005
Main	0.0050	0.0010
Strict	0.0100	0.0020

A witness is stronger when its qualitative interpretation is stable across these threshold variants.

A.17 Breadth and Entropy Diagnostics

Breadth and entropy diagnostics measure multiplicity of macro-labels or macro-paths. They are secondary:

$$H_{\text{macro}} = H(\kappa(s_t)), \quad (190)$$

or, for paths:

$$H_{\text{path}} = H(z_0, z_1, \dots, z_K). \quad (191)$$

These quantities do not define Omega. They are useful only after viable propagation has been established.

A.18 Product Baseline

The product baseline estimates what the propagation statistic would be if the component fields composed independently. Operationally, it uses component-level or independently paired trajectory information to construct a null joint profile without coupled joint transport.

The product baseline tests whether the coupled witness exceeds independent composition.

A.19 Shuffled Baseline

The shuffled baseline preserves marginal trajectory information while breaking coupling-specific joint correlations. It is constructed by re-pairing or shuffling joint trajectory components according to the probe implementation.

The shuffled baseline tests whether a candidate witness depends on real joint coupling structure rather than marginal statistics.

A.20 Independent Calibration

The independent calibration uses $\alpha = 0$ or equivalent uncoupled dynamics to estimate propagation structure without attractive coupling.

This calibration helps distinguish coupling-induced propagation from structure already present in the independent fields.

A.21 Estimator Stability

Probe outputs report bootstrap intervals and estimator diagnostics. A row is treated as more reliable when:

- bootstrap intervals are narrow relative to the effect size;
- null deltas are stable across seeds;
- threshold variants preserve interpretation;
- fragmentation diagnostics remain within bounds;
- component preservation does not collapse.

Estimator stability is not a substitute for theoretical validity, but it prevents overinterpreting finite-sample noise.

B Probe Summary Table

This appendix summarizes the probe sequence that led to the current COM fiber-transport witness. It is not a substitute for the full result folders or running log. Its purpose is to show how the project narrowed from broad candidate ideas to the current quotient-relative toy signature.

B.1 Probe Sequence Interpretation

The probe sequence is best understood as a narrowing process.

Early multifield work suggested that coupling could produce transport-like signals, but those signals were not yet clearly separated from entropy, one-step transport, or quotient artifacts. Probe 09 introduced the robust fiber reachability framing and isolated COM as the first clear multi-step viable propagation candidate.

Probe 10 then tested whether the COM signal was brittle. It was not uniformly robust, but it survived substantial perturbation coverage. This moved COM from “promising local coordinate” to “serious toy witness.”

Probe 11 asked whether COM could be replaced by a simple learned quotient. It could not. The learned methods found partial signal but failed to recover the strong COM coordinate on heldout moderate perturbations.

Probe 12 then separated witness formalization from learned-kappa diagnosis. Probe 12A supported COM as the current formal witness. Probe 12B showed why learned k-means failed. Probe 12C suggested a future direction for learners but did not change the main conclusion.

The resulting project state is:

COM is the current quotient-relative witness. Learned quotient recovery and nearby substrate generalization remain open.

C Repository and Reproducibility Notes

This appendix records the repository structure and reproducibility conventions for the current artifact. The purpose is to make clear which scripts and result folders support the claims in the paper.

Probe	Main Result
Probe 09	Isolated the current candidate object: center-of-mass multi-step viable propagation through certified fibers in the F, T attractive multi-field toy substrate. Distinguished multi-step propagation from local one-step transport.
Probe 10	Tested perturbation robustness of the COM witness. Found that COM viable propagation survived substantial perturbation coverage, with weaknesses concentrated in harder noise and potential-shape variants rather than estimator instability.
Probe 10 Targeted Refinement	Ran higher-coverage COM-only refinement on weaker perturbation families: noise, potential shape, and time discretization. Supported the interpretation that COM propagation is robust but regime-bounded.
Probe 11	Tested whether simple learned predictive k-means quotients could recover viable propagation without being handed COM bins. Found partial learned signal but no strong COM recovery; higher- k learners often fragmented or failed heldout component preservation.
Probe 12A	Audited COM as a formal fiber-transport witness. Reported positive viable propagation index, positive null separation, component preservation, lower-rank erasure control, and tight threshold sensitivity.
Probe 12B	Diagnosed learned-kappa failure modes. Found that simple predictive k-means mostly failed by splitting COM fibers, merging incompatible COM fibers, and inflating small-fiber structure.
Probe 12C	Ran a transition-aware balanced learner smoke test. Improved some anatomy and fragmentation diagnostics, but did not justify scaling learned-kappa methods before formalizing COM.

Table 3: Summary of retained probes leading to the COM fiber-transport witness.

C.1 Repositories

The project currently uses two repositories:

- **Omega-Lab**: executable validation workspace, including probe scripts, generated result folders, and internal diagnostics.
- **Omega**: public-facing repository containing retained compact results, documentation, and selected artifacts.

The current paper is based on retained result sets from Probes 09–12.

C.2 Primary Scripts

The main executable scripts relevant to this artifact are:

- `probe_09_robust_fiber_reachability.py`
- `probe_10_com_viable_propagation_robustness.py`
- `probe_11_learned_predictive_kappa_revised.py`
- `probe_12a_com_formal_object_audit.py`
- `probe_12b_learned_kappa_failure_diagnosis.py`
- `probe_12c_improved_learner_smoke.py`

If Probe 12 is implemented as a combined batch script in the current repository, the corresponding batch script should also be listed here:

- `probe_12_batch_com_audit_learned_diagnosis.py`

C.3 Retained Result Folders

The retained result folders are:

- `probe_09_robust_fiber_reachability_results/`
- `probe_10_com_viable_propagation_robustness_extended_results/`
- `probe_10_com_targeted_fragility_refinement_results/`
- `probe_11_learned_predictive_kappa_revised_results/`
- `probe_12_batch_results/`
- `probe_12a_com_formal_object_audit_results/`
- `probe_12b_learned_kappa_failure_diagnosis_results/`
- `probe_12c_improved_learner_smoke_results/`

Some large generated anatomy tables from Probe 12A are retained locally but not tracked in the public repository because of file-size limits. These include full node, edge, path, and component-projection tables.

C.4 Canonical Result Files

The most important files for reproducing the claims in this artifact are:

- `probe_09_robust_fiber_reachability_results/summary.json`
- `probe_09_robust_fiber_reachability_results/viable_propagation_summary.csv`
- `probe_10_com_viable_propagation_robustness_extended_results/summary.json`
- `probe_10_com_targeted_fragility_refinement_results/summary.json`
- `probe_11_learned_predictive_kappa_revised_results/summary.json`
- `probe_11_learned_predictive_kappa_revised_results/learned_vs_com_comparison.csv`
- `probe_12_batch_results/batch_summary.json`
- `probe_12_batch_results/recommended_next_step.md`
- `probe_12a_com_formal_object_audit_results/summary.json`
- `probe_12a_com_formal_object_audit_results/com_threshold_sensitivity.csv`
- `probe_12a_com_formal_object_audit_results/com_vs_controls_summary.csv`
- `probe_12b_learned_kappa_failure_diagnosis_results/summary.json`
- `probe_12b_learned_kappa_failure_diagnosis_results/learned_vs_com_anatomy_summary.csv`
- `probe_12c_improved_learner_smoke_results/summary.json`

C.5 Compute Environment

The current validation runs were performed on midtier consumer hardware using CPU multiprocessing. The standard calibrated worker count is:

$$N_{\text{workers}} = 18. \quad (192)$$

Probe 12 completed at fallback scale with:

$$N_{\text{traj}} = 3000, \quad N_{\text{seeds}} = 100, \quad N_{\text{bootstrap}} = 300. \quad (193)$$

Earlier Probe 10 robustness runs used larger or targeted settings as described in the corresponding summaries.

C.6 Reproducibility Convention

Each probe should write at least:

- `summary.json`
- primary metric CSVs;
- bootstrap or uncertainty tables;

- estimator diagnostics;
- plots where useful;
- a final console summary.

Probe outputs should distinguish:

- primary propagation metrics;
- secondary entropy or breadth diagnostics;
- component-preservation diagnostics;
- nonfragmentation diagnostics;
- null-baseline deltas;
- estimator warnings.

C.7 Public Caveat

The retained public results are selected to communicate the current state of the program. Older exploratory artifacts remain useful internally but are not necessary for understanding the current COM witness. The absence of older outputs from the public tree should not be interpreted as evidence that they do not exist; rather, they are summarized in the project manual and running log.

The current public interpretation is:

COM-like multi-step viable propagation through certified fibers in the F, T attractive multifold toy substrate is the current executable candidate.

This interpretation remains bounded to the toy substrate and the current probe sequence.